# MAKING THE RIGHT CONNECTIONS: IMPROVING THE COMPREHENSION OF STRUGGLING READERS

**GRANT# R324A110046**
**ANNUAL REPORT, September 2012-August 2013**

## I. ACCOMPLISHMENTS

### A. Major Goals of the Project

The major goals of this project, as specified in the performance agreement, are to produce one or more fully developed questioning intervention(s) aimed at improving the reading comprehension of students at risk or identified as having disabilities. Intended products include: (a) one or more prototypes that are feasible and promising, with manuals; (b) one or more observational checklists to be used to document fidelity; and (c) training materials to prepare school-based personnel for implementation. We also aim to identify for whom (subgroups of struggling readers) and under what conditions (type and timing of questions, intensity of intervention) the intervention is most promising.

### B. What was Accomplished under these Goals in Year 2

In Year 2, we completed or are in the process of completing the following activities to meet the above project goals: (1) Revise core intervention questions, materials, and procedures (complete); (2) Identify struggling readers to participate in pilot studies (complete); (3) Conduct trial runs and a pilot study to examine feasibility and student outcomes (complete); (4) Analyze and interpret data from Year 2 study activities (in process); and (5) Refine interventions based on results of Year 2 study activities (in process). These activities fall into three main categories: Identifying and Describing Struggling Readers, Text Selection and Development, and Intervention Development. Figure 1 provides details of activities under each category. Specific methods and results for each category are described in detail below.

**Figure 1. Year 2 Timeline**

<table>
<tr><th></th><th>Identifying and Describing Struggling Readers</th><th>Revising Existing and Selecting New Texts; Developing & Refining Questions</th><th>Developing Intervention</th></tr>
<tr><td>Sep-12</td><td>Screening Phase 1</td><td rowspan="10">Cycle of Identifying Potential Texts, Transcribing, Determining Text Ease/ Readability, Developing & Reviewing Questions and Possible Responses OR refining existing texts/questions</td><td rowspan="2">Round 6: In-House Development</td></tr>
<tr><td>Oct-12</td><td rowspan="2">Screening Phase 2 & Think-Aloud Administration</td></tr>
<tr><td>Nov-12</td><td>Round 7: Focus Groups & Expert Review</td></tr>
<tr><td>Dec-12</td><td rowspan="6">Think-Aloud Coding</td><td>Round 8: Fall Trial Run</td></tr>
<tr><td>Jan-13</td><td rowspan="3">Round 9: Spring Pilot</td></tr>
<tr><td>Feb-13</td></tr>
<tr><td>Mar-13</td></tr>
<tr><td>Apr-13</td><td>Scoring and data entry</td></tr>
<tr><td>May-13</td><td>Round 10: Spring Trial Run</td></tr>
<tr><td>Jun-13</td><td>Subgroup ID</td><td rowspan="3">Revision of Texts and Questions</td><td>Spring Pilot recall coding</td></tr>
<tr><td>Jul-13</td><td>Data analysis</td><td>Data analysis</td></tr>
<tr><td>Aug-13</td><td></td><td>Round 11: In-House Revision of Intervention</td></tr>
</table>

## IDENTIFYING AND DESCRIBING STRUGGLING READERS

The purpose of identifying and describing struggling readers was to (a) identify students struggling primarily with comprehension, and thus eligible for the intervention to be developed and (b) identify subgroups of struggling comprehenders. Specific research questions include:

(1) Of a diverse sample of fourth-grade students, which students are struggling primarily with reading comprehension (not decoding/word recognition or fluency skills)?
(2) Of those students identified as struggling comprehenders, do subgroups of comprehenders emerge, based on responses to a think-aloud task?

## Method

### Participants

**Districts and schools**. Approval to conduct this research was obtained from two districts' Departments of Research, Evaluation & Assessment (REA) prior to recruiting teachers and students. District 1 is a large urban district serving about 32,263 K-12 students in 70 schools at the time of this study (2012-13 school year). District 2 is a first-ring suburb of a large urban area serving 10,622 K-12 students in 17 schools at the time of this study. A total of seven schools across the two districts agreed to participate. District and school demographic information for the study year (2012-13) is summarized in Table 1.

**Table 1.** ***Participating District and School Demographic Information***

| | Total Enrollment | Students of Color (%) | English Learners (%) | Free/Reduced Lunch (%) | Special Ed (%) |
|---|---|---|---|---|---|
| District 1 | 32,263 | 68 | 21 | 66 | 19 |
| School 1 | 407 | 88 | 27 | 83 | 10 |
| School 2 | 715 | 17 | 2 | 9 | 8 |
| School 3 | 617 | 94 | 60 | 95 | 17 |
| School 4 | 636 | 22 | 1 | 17 | 10 |
| District 2 | 10,622 | 43 | 12 | 39 | 15 |
| School 5 | 580 | 42 | 18 | 34 | 12 |
| School 6 | 525 | 41 | 19 | 35 | 14 |
| School 7 | 544 | 86 | 44 | 80 | 9 |

At the end of Year 1, we invited all participating teachers to participate in Year 2. Most teachers at least tentatively agreed (except for teachers who knew they were leaving their current position or grade level). In fall of Year 2, we followed up with all teachers. In cases where Year 1 teachers were no longer available, new teachers were brought on board (e.g., in some schools, there were new 4$^{th}$-grade teachers in an already-participating school; or, one Year 1 teacher moved to a new school but continued participating, so we invited his new colleague to join us), we met with grade-level teams (and in the new school, the building principal) to provide an overview of the study and to address questions. In some cases, reading specialists and/or special education teachers also attended the meetings. New teachers either expressed interest on the spot or within approximately one week. We then contacted school principals to confirm their approval of the teachers' participation. Table 2 provides a summary of teachers who returned, withdrew, or enrolled as new teachers.

**Table 2.** ***Summary of New and Participating Teachers***

| | Returning | Withdrawn | Newly Enrolled |
|---|---|---|---|
| District 1 | | | |
| School 1 | 1 gr. 4 (moved from school 3) | 0 (new school) | 1 gr. 4<br>1 ELL specialist |
| School 2 | 2 gr. 4 | 1 gr. 4 (moved to gr. 5)<br>1 coach (no longer at school) | 2 gr. 4 |
| School 3 | 1 gr. 4<br>1 reading coach | 1 reading resource teacher (retired),<br>1 ELL teacher (no longer at school) | 1 special ed teacher<br>1 ELL specialist |
| School 4 | 4 gr. 4 | 1 reading coach | 1 special ed teacher |
| School 5 | 0 | 3 (school opted out) | 0 |
| District 2 | | | |
| School 5 | 4 gr. 4 | 2 resource teachers (no longer at school) | 2 intervention specialists<br>1 special ed teacher |
| School 6 | 0 | 4 (moved to gr. 5) | 4 gr. 4 |
| School 7 | 4 gr. 4, 1 ELL | 0 | 0 |
| TOTALS | 18 | 14 | 14 |

**Teachers**. Thirty-two teachers participated, including 23 fourth-grade classroom teachers and 9 specialists. Specialists described their positions and job responsibilities as follows: 2 intervention specialists (taught reading/implemented guided reading and provided reading intervention to at risk or at some risk in reading), 1 reading coach (provided professional development to literacy teachers and implemented RTI literacy model), 3 special education teachers (1 multi-categorical teacher, 1 LD teacher, and 1 SLD/EBD teacher; maintained caseloads, provided intervention and RTI support), 3 EL teachers (provided reading & writing support in collaborative and pull-out models). Teacher-reported demographic data are in Table 3.

**Students**. All fourth-grade students in participating classrooms were invited to participate in the study. Research staff provided brief (5-10 min) presentations of the study to each participating class, and distributed consent forms to all students to take home and have signed by their parents. Students received gel pens when they returned the signed consent forms (whether the parents indicated 'yes' or 'no' to participation). Table 4 shows the distribution and return rates of parental consent forms in the two districts.

## Measures

**Screening measures**. Screening occurred in two phases. Phase I screening measures included CBM maze tests. Phase II screening measures included CBM oral reading and district tests. Table 5 shows intercorrelations among screening measures.

**Curriculum-Based Measurement (CBM)—Maze Task**. The CBM maze task (e.g., Fuchs & Fuchs, 1992) is designed to be a brief assessment of overall reading proficiency. The maze is group administered, and takes about 10-15 min to administer. In this study, students read two passages in which every seventh word was deleted and replaced with three word choices, only one of which fit in the sentence. Students completed two practice sentences, and then read each passage silently for 2.5 min, selecting the words that best fit in the sentences. The score was the average correct maze selections made on the two passages. Alternate-form reliability

coefficients have been reported to range from r = .83 to .98, and correlations between maze selection scores and other reading measures typically range from r = .60 to .80 for elementary-aged students (Wayman, Wallace, Wiley, Ticha, & Espin, 2007). In the screening sample for this study (n = 461), test-retest reliability for the maze was r = .86, p < .001.

**Table 3.** ***Teacher-Reported Demographics***

| | Classroom teachers (*n*=23) | | Specialists (*n*=9) | |
|---|---|---|---|---|
| | n | % | n | % |
| Sex (female) | 17 | 73.9 | 9 | 100.0 |
| Race | | | | |
| White | 17 | 73.9 | 8 | 88.9 |
| Hispanic | 1 | 4.3 | 1 | 11.1 |
| American Indian | 1 | 4.3 | 0 | 0.0 |
| Other | 3 | 13.0 | 0 | 0.0 |
| Age | | | | |
| 20-29 | 3 | 13.0 | 2 | 22.2 |
| 30-39 | 9 | 39.1 | 2 | 22.2 |
| 40-49 | 5 | 21.7 | 2 | 22.2 |
| 50-59 | 5 | 21.7 | 1 | 11.1 |
| 60 + | 1 | 4.3 | 2 | 22.2 |
| Highest Degree | | | | |
| Bachelor's | 4 | 17.4 | 3 | 33.3 |
| Master's | 5 | 21.7 | 2 | 22.2 |
| Master's + | 14 | 60.9 | 4 | 44.4 |
| | *M* (range) | *SD* | *M* (range) | *SD* |
| Years in current position | 7.14 (1-20) | 5.4 | 8.75 (1-25) | 10.0 |
| Years in general education | 13.73 (2-29) | 7.5 | 14.17 (1-40) | 14.2 |
| Years teaching Grade 4 | 9 (1-29) | 7.9 | 7.33 (1-18) | 6.5 |
| Years teaching special ed. | 7.33 (1-13) | 6.0 | 11.67 (0-23) | 11.5 |
| Years as a reading specialist | 4.67 (4-6) | 1.2 | 6.5 (1-12) | 7.8 |
| Students in class/caseload | 27.48 (21-53) | 8.1 | 74.22 (21-400) | 122.8 |

**Table 4.** ***Distribution and Return Rates of Consent Forms***

| | Distributed | Returned Indicating 'Yes' or 'No' to Participation | | | Return Rate |
|---|---|---|---|---|---|
| | | Yes | No | Total | |
| District 1 | 262 | 209 | 32 | 241 | 87% |
| District 2 | 346 | 252 | 48 | 300 | 84% |
| Total | 608 | 461 | 80 | 541 | 85% |

**District tests**. Both districts used the Measures of Academic Progress (MAP), which was developed by the Northwest Evaluation Association and assesses students' reading achievement and growth from late elementary through high school. Reading achievement is assessed in Minnesota in the areas of "Literature," "Informational Text," "Foundational Skills and Vocabulary" and "Language Use and Structure." These areas are aligned with district and state reading standards. All MAP assessments are computerized and presented adaptively, adjusting in difficulty based on each student's responses to test questions. They commonly consist of more than 40 questions. Assessments are provided in a survey structure with content area goal scores supplementing an overall score. For the purposes of this study, we recorded the Normal Curve Equivalent (NCE) and RIT score, which allows the values of achievement levels and the values of items to be on the same scale. The reported stratified, randomly parallel form reliability for 4th grade is .80 (Northwest Evaluation Association, 2011).

**CBM oral reading**. The CBM oral reading task (Deno, 1985) consists of individually administered grade-level passages, which students read aloud for 1 min. As the student reads, the examiner marks insertions, omissions, substitutions, and mispronunciations not caused by speech-related problems or dialects. Errors are not corrected and no decoding assistance is given. In this study, the student read two passages; the score is the mean number of words read correctly in one min. For this study, we used two Formative Assessment System for Teachers (FAST; Christ, Ardoin, Eckert, & White, 2010a) 4th-grade-level CBM passages. Criterion validity for the 4th-grade passages with other reading measures (DIBELS and TOSREC) has been reported as r = .95 (DIBELS) and .74 (TOSREC), and classification accuracy for identifying students at risk has been reported as strong (Christ, Ardoin, Eckert, & White, 2010b). In the screening sample for this study (n = 135), test-retest reliability was r = .87, p < .001.

**Table 5.** ***Intercorrelations among Screening Measures***

| | 1. Maze | 2. MAP | 3. CBM Oral Reading |
|---|---|---|---|
| 1. Maze | | | |
| 2. MAP | .64* (*n* = 90) | | |
| 3. CBM Oral Reading | .55* (*n* = 135) | .62* (*n* =58) | . |

*Note.* * Significant correlation coefficients (p < .001). MAP = Measures of Academic Progress. *N*s vary because MAP data in District 1 were only available for students who were potentially eligible based on Maze scores, and CBM Oral Reading data were collected only from students in Screening Phase II.

**Think-aloud task**. The think-aloud task was used to identify possible subgroups of struggling comprehenders. As in previous related studies (e.g., McMaster et al., 2012; Rapp et al., 2007), we used a think-aloud task adapted from Ericsson & Simon (1993), which included a practice text ("Jimmy and the New Bike") adapted from Magliano, Trabasso, & Graesser, 1999, and two stories ("Brian's Magical Skill" and "Candle Crafts for School") and two informational texts ("Living Underground: The Life of a Mole" and "Mountain Heights") adapted from van den Broek et al., 2001). Each sentence from each text was typed on 5" X 6" cards. Think-aloud texts are provided in Appendix A.

The examiner demonstrated the think-aloud task with the first half of the practice text by giving specific examples of all possible types of responses (e.g., elaborative inferences, paraphrases, predictions). Then, the student practiced thinking aloud with the remaining sentences in the practice text. The student read one sentence at a time and then told the examiner

what he or she was thinking. After each text, the examiner asked the student two yes/no questions about the text to check for general comprehension. The student received no decoding help, but received non-leading prompts such as, "What are you thinking now?" if he or she forgot to think aloud.

**Motivation measure**. In January 2013, before intervention began, we administered a Motivation for Reading Questionnaire (MRQ; Wigfield & Guthrie, 1997) to all students identified as eligible for the intervention. The MRQ is used to more completely describe the struggling readers, as well as to determine if motivational factors moderate students' comprehension of different text genres under varying intervention conditions.

The MRQ (Wigfield & Guthrie, 1997) assesses 11 aspects of reading motivation: reading efficacy, reading curiosity, reading involvement, importance of reading, reading challenge, reading work avoidance, competition in reading, recognition of reading, reading for grades, social reasons for reading, and compliance. Only the first four aspects were assessed as part of this study, on the recommendation of Guthrie (personal communication, January 2012).

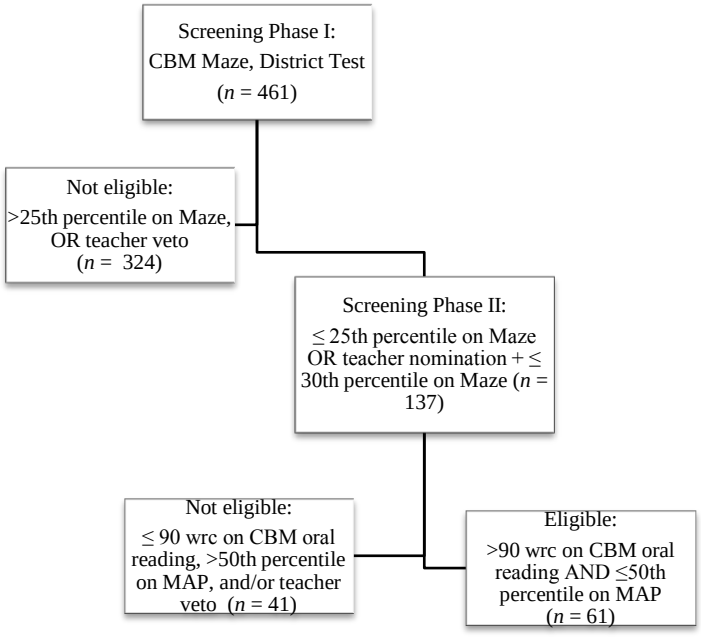
The MRQ is an independent, group-administered, self-report measure of reading motivation. The full 11 construct MRQ contains 53 items and takes 15 to 20 minutes to administer. The 4-construct version used in this study included 17 items and took approximately 5 min to administer. As a group, students answered two practice questions, which asked them to identify to their level of agreement with a statement (e.g., "I like ice cream") using a four point rating scale (1 = very different than me to 4 = a lot like me). Students then completed the 17 MRQ items independently, while the teacher read each statement out loud to the group.

## Procedures

**Screening**. To determine eligibility for the intervention, all consented students (n = 461) were screened in two phases (see Figure 2). In Screening Phase I, conducted in September 2012, research staff administered two CBM maze probes. Students whose average scores from the two CBM maze probes were at or below the 25th percentile were identified as potentially eligible for the study and proceeded to Screening Phase II. The bottom 25th percentile of the two districts together was 14.5 correct maze choices (cmc) (District 1 = 15.5, District 2 = 13.5); we used the full sample percentile.

Those students identified as eligible for Screening Phase II, conducted in October-November, 2012, were individually administered two CBM oral reading probes. Those whose mean score was less than or equal to 90 words read correctly (wrc) in one min, or whose teachers felt they would not benefit from intervention due to poor decoding or English language skills, were identified as ineligible for intervention ($n$ = 41; 9% of sample), because their reading difficulties were likely at least partly due to decoding or word recognition problems (Christ et al., 2010b). Those who scored greater than 90 wrc in one min (but did not score above the 50$^{th}$ percentile on the district MAP test) were identified as eligible for the intervention ($n$ = 61; 13%).

**Figure 2.** ***Participant Flow through Screening Process***




**Think-aloud administration and coding**. Students identified as eligible for the intervention based on CBM oral reading scores were also administered think-aloud tasks to identify subgroups of struggling readers. The think-aloud task was conducted using a script within the "CBM Oral Reading and Think-Aloud" directions. The Discourse Project Coordinator trained GRAs to administer the think-aloud task by: (1) modeling the procedure, (2) observing each GRA walk through the procedure according to the script, using the story cards, and audio-recording the session, and (3) verifying proficiency of administration by the GRA after a period of independent practice.

Think-aloud tasks were administered individually during Screening Phase II, using two narrative and two informational texts. GRAs were instructed to only administer the think-aloud task if the student had scored at least 90 words read correctly on at least one of the CBM oral reading probes. The entire session was audio-recorded and transcribed.

Undergraduate Research Assistants (URAs) and GRAs transcribed verbatim the think-aloud audio-recordings according to a written procedure. Twenty percent of the transcripts, including at least one for each URA transcriber, were verified for accuracy by the Discourse Project Coordinator or a GRA who was experienced and proficient in transcription.

Think-aloud transcripts were then parsed into thought units, which represent a complete idea with subject and verb, and possibly subordinate clause. Each parsed thought unit was categorized based on a coding scheme adapted from previous research (Linderholm & van den Broek, 2002) designed to identify the cognitive processes in which readers engaged during reading. The primary variables coded included:

- Elaborative inferences (readers' attempts to explain the contents of the current sentence using background knowledge. Coded as valid when explanations were constrained by the text, plausible and accurate, or invalid when explanations were not constrained by the text, were implausible or reflected a misunderstanding or other error)

- Predictive inferences (readers' anticipations of what would occur next in the text, also coded as valid or invalid based on whether the predictions were constrained by the text, were plausible or involved error)
- Connecting inferences (readers' explanations of the current sentence by connecting it to the immediately preceding sentence)
- Reinstatement inferences (readers' explanations of the current sentence on the basis of prior text information not in the immediately preceding sentence)
- Associations (readers' comments about the current sentence that draw upon background knowledge as an aside
- Paraphrases (readers' rephrasing of the current sentence in their own words)
- Text repetitions (readers' verbatim reiterations of the current sentence)
- Metacognitive comments (readers' reflections on their understanding of text.)
- Evaluative comments (readers' opinions about the text.)
- Affective responses (readers' reflections of their own emotions in response to the current sentence)
- Questions (readers' questions asked about the content of the current sentence).
- Other (readers' statements that cannot be categorized often because of inaudible utterances)

The coding scheme used in Year 2 reflect the infrequent codes that were collapsed in Year 1; that is, Metacognitive Comment encompassed comments that reflected agreement, disagreement or lack of background knowledge, and Evaluative Comments encompassed evaluations of either content or function of the text. Low frequency codes were collapsed (Paraphrases and text repetitions were combined into one "text paraphrase" code; metacognitive, evaluative, and affective responses were combined into one "comment" code). "Other" codes were excluded from analysis. Frequencies of each code on narrative, informational and all texts are in Table 6.

**Table 6.** ***Frequencies of Think-Aloud Codes***

| CODE | NAME OF CATEGORY | Narrative | Informational | Total |
|---|---|---|---|---|
| EII | Elaborative Inference Invalid | 96 | 157 | 253 |
| EIV | Elaborative Inference Valid | 918 | 1105 | 2023 |
| PII | Predictive Inference Invalid | 28 | 3 | 31 |
| PIV | Predictive Inference Valid | 634 | 214 | 840 |
| CI | Connecting Inference | 222 | 47 | 269 |
| RI | Reinstatement Inference | 645 | 177 | 822 |
| A | Association | 150 | 450 | 600 |
| TP | Text Paraphrase | 865 | 559 | 1424 |
| CM | Comment | 87 | 102 | 189 |
| Q | Question | 16 | 8 | 24 |
| O | Other | 9 | 4 | 13 |

URAs coded the transcribed think-alouds. Training and practice proceeded as follows:

1. ***Training sessions***. Based on the Year 1 training provided by think-aloud expert Dr. Sarah Carlson, the Discourse Project Coordinator and advanced GRA taught the Year 2 URAs the purpose of the think-aloud measure, definitions of codes, and rules and procedures for coding. They modeled coding and guided practice in coding.
2. ***Group coding practice***. Project coordinator, advanced GRA and two or more URAs practiced applying the think-aloud coding rules and procedures to the previous year's think-aloud transcripts. The coding practice was in-depth and continued even as the Year 2 think-aloud tasks were being transcribed.
3. ***Paired coding practice***. Pairs of URAs coded transcripts together until they were comfortable with their level of agreement. Then they coded the same protocol independently before reaching consensus on codes for that protocol. When pair-wise IRA on the consensus protocol reached 75%, the pair proceeded to individual coding.
4. ***Individual coding with every third transcript subjected to consensus coding***. Pairs coded transcripts in sets of five; three each were coded individually, and one transcript was reviewed for consensus. If pair-wise IRA on the consensus protocol fell below 75%, the pair returned to partner coding.
5. ***Third party IRA for pairs of coders***. The Discourse Project Coordinator or advanced GRA coded If IRA averaged less than 75%, all subsequent transcripts were recoded.

Because much of the training and practice coding Year 1 protocols occurred before a 4-week holiday and semester break, when coding resumed after the break the URAs coded Year 2 protocols in groups and in pairs to recalibrate their coding decisions.

The discourse project coordinator and advanced GRA coded protocols until their pairwise IRA exceeded 75%, and then they established IRA with individual URAs. Narrative think-alouds were coded separately from expository think-alouds, so the 61 subjects' think-alouds were coded as 122 protocols. Twenty-two protocols were coded by practice groups, 38 were coded by pairs and 62 were coded by individuals. Thirty-one percent of the protocols were subjected to pairwise or third party IRA. Overall pair-wise IRA ranged from 74-88%; on narratives it averaged 85%; and on informational texts it averaged 87%.

Kappa was calculated between discourse project coordinator or advanced GRA and individual URAs and between pairs of URAs. Kappa ranged from .58 to .91. The URA with the lower Kappa had undergone further practice coding to increase IRA.

**MRQ test administration and scoring**. The MRQ was administered individually in late January 2013 as part of the pretest. Examiners were trained to provide the standardized instructions, and asked to monitor students closely to make sure they answered all of the questions.

## Results

**Research Question 1:**

Of a diverse sample of fourth-grade students, which students are struggling primarily with reading comprehension (not decoding/word recognition or fluency skills)? This question was addressed through the screening process described above. Table 7 shows descriptive data obtained from Screening Phases I and II as well as for students identified as eligible for intervention.

**Table 7.** ***Descriptive Statistics from Screening***

| | Screening Phases I and II | | | | |
|---|---|---|---|---|---|
| | *n* | Mean | *SD* | Min | Max |
| Maze | | | | | |
| Phase I | 461 | 19.10 | 6.91 | 0 | 38 |
| Phase II | 137 | 11.80 | 3.14 | 2 | 16 |
| | Eligible for Intervention | | | | |
| | *n* | Mean | *SD* | Min | Max |
| Maze | 61 | 11.77 | 2.51 | 4 | 15 |
| MAP SS | | 190.20 | 7.22 | 168 | 202 |
| MAP Percentile | | 27.90 | 14.19 | 1 | 58 |
| Oral Reading | | 114.36 | 16.34 | 90 | 161 |

**Research Question 2:**

Of those students identified as struggling comprehenders, do subgroups of comprehenders emerge, based on responses to a think-aloud task?

To identify possible subgroups of struggling readers, think-aloud data were submitted to a cluster analysis using the k-means cluster approach for both the narrative and informational texts. Given that in previous studies, a two-group solution best fit the data, we used a two-group solution. Two expected groups emerged: one that was primarily distinguished by paraphrasing ('paraphrasers,' $n = 15$ for narrative and 21 for informational) and one that was primarily distinguished by elaborating ('elaborators,' $n = 46$ and 40 for narrative and informational). The two groups emerged in the expected proportions based on previous research (with more Elaborators than Paraphrasers; e.g., McMaster et al., 2012; Rapp et al., 2007; van den Broek et al., 2006). Further, a majority of students (90%) were identified as being in the same subgroup for both the narrative and informational think-aloud tasks. This finding is noteworthy given that a goal of this project was to determine whether the think-aloud approach yielded consistent subgroups using informational text as it did using narrative text, and this replicates Year 1 findings (when 94% were identified as being in the same subgroup for both narrative and informational).

**Table 8.** ***Proportions of Think-Aloud Statements made by Three Subgroups***

| | Narrative Think-Aloud | | | | | | Informational Think-Aloud | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *n* | Mean | *SD* | *F* (1, 59) | *p* | | *n* | Mean | *SD* | *F* (1, 59) | *p* | |
| Associations | | | | 5.77 | .019 | E > P | | | | 8.21 | .006 | E > P |
| Elaborators | 46 | 0.10 | 0.14 | | | | 40 | 0.07 | 0.10 | | | |
| Paraphrasers | 15 | 0.01 | 0.02 | | | | 21 | 0.00 | 0.01 | | | |
| Total | 61 | 0.07 | 0.12 | | | | 61 | 0.04 | 0.09 | | | |
| Connecting Inferences | | | | 0.46 | .502 | ns | | | | 0.04 | .851 | ns |
| Elaborators | 46 | 0.04 | 0.03 | | | | 40 | 0.04 | 0.04 | | | |
| Paraphrasers | 15 | 0.04 | 0.02 | | | | 21 | 0.04 | 0.04 | | | |
| Total | 61 | 0.04 | 0.03 | | | | 61 | 0.04 | 0.04 | | | |
| Reinstatements | | | | 13.66 | <.001 | E > P | | | | 10.42 | .002 | E > P |
| Elaborators | 46 | 0.16 | 0.09 | | | | 40 | 0.13 | 0.06 | | | |
| Paraphrasers | 15 | 0.07 | 0.05 | | | | 21 | 0.07 | 0.07 | | | |
| Total | 61 | 0.14 | 0.09 | | | | 61 | 0.11 | 0.07 | | | |
| Elaborative Inferences - Invalid | | | | 6.83 | .011 | P > E | | | | 2.28 | .136 | ns |
| Elaborators | 46 | 0.04 | 0.03 | | | | 40 | 0.04 | 0.04 | | | |
| Paraphrasers | 15 | 0.07 | 0.04 | | | | 21 | 0.02 | 0.03 | | | |
| Total | 61 | 0.05 | 0.04 | | | | 61 | 0.03 | 0.04 | | | |
| Elaborative Inferences - Valid | | | | 13.62 | <.001 | E > P | | | | 53.35 | <.001 | E > P |
| Elaborators | 46 | 0.35 | 0.13 | | | | 40 | 0.40 | 0.13 | | | |
| Paraphrasers | 15 | 0.21 | 0.12 | | | | 21 | 0.17 | 0.09 | | | |
| Total | 61 | 0.31 | 0.14 | | | | 61 | 0.32 | 0.16 | | | |
| Predictive Inferences - Invalid | | | | 1.25 | .268 | ns | | | | 1.23 | .272 | ns |
| Elaborators | 46 | 0.01 | 0.02 | | | | 40 | 0.01 | 0.02 | | | |
| Paraphrasers | 15 | 0.00 | 0.00 | | | | 21 | 0.00 | 0.00 | | | |
| Total | 61 | 0.00 | 0.02 | | | | 61 | 0.00 | 0.02 | | | |
| Predictive Inferences - Valid | | | | 12.16 | .001 | E > P | | | | 14.86 | <.001 | E > P |
| Elaborators | 46 | 0.16 | 0.11 | | | | 40 | 0.15 | 0.10 | | | |
| Paraphrasers | 15 | 0.05 | 0.05 | | | | 21 | 0.06 | 0.03 | | | |
| Total | 61 | 0.13 | 0.11 | | | | 61 | 0.11 | 0.09 | | | |
| Metacognitive Comments | | | | 1.21 | .276 | ns | | | | 2.16 | .147 | ns |
| Elaborators | 46 | 0.03 | | | | | 40 | 0.03 | 0.06 | | | |
| Paraphrasers | 15 | 0.01 | | | | | 21 | 0.01 | 0.02 | | | |
| Total | 61 | 0.03 | | | | | 61 | 0.02 | 0.05 | | | |
| Text Repetitions/Paraphrases | | | | 187.56 | <.001 | P > E | | | | 203.77 | <.001 | P > E |
| Elaborators | 46 | 0.11 | 0.08 | | | | 40 | 0.14 | 0.11 | | | |
| Paraphrasers | 15 | 0.53 | 0.15 | | | | 21 | 0.62 | 0.15 | | | |
| Total | 61 | 0.21 | 0.21 | | | | 61 | 0.30 | 0.26 | | | |
| Questions | | | | 3.72 | .059 | ns | | | | 0.84 | .364 | ns |
| Elaborators | 46 | 0.01 | 0.01 | | | | 40 | 0.00 | 0.02 | | | |
| Paraphrasers | 15 | 0.00 | 0.00 | | | | 21 | 0.00 | 0.00 | | | |
| Total | 61 | 0.00 | 0.01 | | | | 61 | 0.00 | 0.02 | | | |
| Other | | | | 0.77 | .383 | ns | | | | 1.02 | .317 | ns |
| Elaborators | 46 | 0.00 | 0.01 | | | | 40 | 0.00 | 0.01 | | | |
| Paraphrasers | 15 | 0.00 | 0.01 | | | | 21 | 0.00 | 0.00 | | | |
| Total | 61 | 0.00 | 0.01 | | | | 61 | 0.00 | 0.01 | | | |

## REVISING EXISTING/SELECTING NEW TEXTS AND DEVELOPING/REFINING QUESTIONS

The purpose of revising existing and selecting new texts and developing/refining questions was to refine and extend our pool of stories and informational texts that fit with the schools' reading instructional programs, were at an appropriate level for the struggling readers in the study, and were amenable to the causal questioning approach under investigation; and to develop and refine causal questions to be used during intervention.

### Method

Year 2 texts included a subset of stories used during Year 1, as well as one additional informational text. Existing questions were revised/refined, and new questions developed, with feedback from co-PI van den Broek and Discourse Expert, Dr. David Rapp of Northwestern University.

### Measures

**Coh-Metrix Text Easability Assessor**. As in Year 1, we used the web-based Coh-Metrix Text Easability Assessor (http://141.225.42.101/cohmetrixgates/) to establish the readability as well as other important dimensions of each text. The Text Easability Assessor provides percentile scores on five characteristics of text extracted from a wider range of linguistic features calculated by Coh-Metrix: Narrativity, Syntactic Simplicity, Word Concreteness, Referential Cohesion, and Deep Cohesion. The five text easability scores were extracted from a wide range of linguistic features calculated by Coh-Metrix, and are recommended for use given that they have been found to account for most of the variance in text across grade levels and text types (Graesser, McNamara, & Kulikowich, 2011). Graesser et al. (2011) define these features as follows:

- ***Narrativity***: "The extent to which the text conveys a story, a procedure, or a sequence of episodes of actions and events with animate beings" (p. 228).
- ***Syntactic simplicity***: "Scores are higher when sentences have fewer words and simpler, more familiar syntactic structures" (p. 230).
- ***Word concreteness***: "Scores are higher when a higher percentage of content words are concrete, are meaningful, and evoke mental images—as opposed to being abstract" (p. 230).
- ***Referential cohesion***: "The extent to which explicit content words and ideas in the text are connected with each other as the text unfolds" (p. 230).
- ***Causal cohesion***: "Scores are higher to the extent that clauses and sentences in the text are linked with causal and intentional (goal-oriented) connectives" (p. 230).

**Question development rubric**. Project staff and discourse experts used the question development rubrics (Appendix B), developed in Year 1, that prompted them to read the text; make sure it was coherent, did not require excessive background knowledge, and was age-appropriate; identify the goal structure; identify places where a causal inference would likely be needed; and generate questions designed to prompt such inferences. Guidelines were also provided regarding the wording and formatting of questions.

## Procedures

**Text selection**. Texts selected for the intervention had to meet the following criteria:

(1) They had to fit with the types of texts used in participating schools' instructional programs in terms of difficulty level, genre, content, age-appropriateness, and interest level (but did not duplicate specific texts);
(2) They had to be sufficiently difficult but not too difficult for students struggling primarily with reading comprehension (generally, in the O-R range as identified by teachers in Year 1 of the project);
(3) They had to be short enough to be read in 3-4 sessions; and
(4) They had to have a structure that lent itself to the causal questioning approach to be used in the intervention (e.g., a clear goal with resolution for narrative text; a cause/effect or problem/solution structure for informational text).

In Year 1 of the project, we had identified 6 stories (divided into segments for 18 sessions) and two informational texts (divided into segments for 6 sessions). For Year 2, we determined that we would need two stories for a Trial Run (divided into segments for 5 sessions), and 3 stories and 3 informational texts for the pilot study (divided into segments for 9 sessions of each text type). Thus, our text selection process involved identifying a subset of existing stories and searching for a third informational text.

**Stories**. We determined which stories to use by first asking project staff who had served as tutors in the previous year to rank-order the existing texts from their most to least favorite, based on what they had perceived to work well with the students in terms of appropriate interest, challenge, and fit with the questioning approach. We also examined student satisfaction data from Year 1, in which participating students were asked to rate each text on a four point Likert-type scale indicating how much they liked each text.

Tutor rankings and student ratings converged on the following texts: "Androcles and the Lion" (rated only by tutors because students did not read this text in the Year 1 pilot study), "A Different Aladdin," "Little Singing Frog," and "The Emperor and the Peasant Boy." These texts are somewhat similar in genre (they are derivatives or adaptations of fables and folk tales) but very different in content. All texts had been identified in Year 1 as appropriate in terms of difficulty level, genre, content, age-appropriateness, and interest level for struggling fourth-grade readers by reading specialists and discourse experts. All texts could be divided into two or three segments, and had causal structures.

**Informational texts**. In Year 1, we identified two informational texts for the Spring 2012 Trial Run: "Jane Goodall and the Chimpanzees," and "Olympic Dreams." We searched for a third informational text using the Scholastic Book Wizard (www.scholastic.com/bookwizard/), in which the desired reading level and genre are selected, and a list of books meeting these criteria is generated. We searched for informational texts in the O-R range, scanned titles and book descriptions, selected promising titles, and uploaded them into an excel file. Project staff then located the texts in local public libraries, and brought them to the lab for review. The project staff then reviewed the texts based on the following criteria: (1) age-appropriateness including vocabulary and content; (2) interest level for struggling fourth graders; (3) text length, which needed to be long enough to be used across 2 – 3 days; (4) whether causal questions could be generated for the text.

Texts that passed this review process were transcribed and entered into the CohMetrix Text Easability Assessor. Unfortunately, most of the texts were either too difficult (above a 5th-grade level) and/or did not lend themselves to the questioning approach (we had encountered a similar problem in Year 1). We found that many informational texts either did not follow a causal structure (which was not surprising), or, if they did, they were so transparent such that no text-based causal or explanatory inferences needed to be generated (or could be generated) to comprehend the text. Most of the texts also required substantial background knowledge for sufficient comprehension (also not surprising)—which is important, but would require a different type of intervention than the one under investigation.

After much discussion about these problems, we determined that the most appropriate "informational texts" for this study were best described as "narrative nonfiction," such as biographical and other types of historical texts. One text identified in the process described above met this criterion: "Cesar Chavez." Thus, the three texts identified for Year 2 were "Cesar Chavez," "Jane Goodall and the Chimpanzees," and "Olympic Dreams." These texts are somewhat similar in genre (they are all biographical) but very different in content (one about a social activist, one about a scientist, and one a collection of short biographies of Olympic athletes). All texts were determined by reading specialists and discourse experts to be appropriate in terms of difficulty level, genre, content, age-appropriateness, and interest level for struggling fourth-grade readers. All texts could be divided into at least two or three segments, and had causal structures.

**Question and response development/refinement**. Two GRAs who were trained to develop questions and possible responses for the intervention texts using the Question Development Rubric (described above) developed questions for the new text ("Cesar Chavez"). For all of the existing texts and the new text, we then followed four steps to refine the questions and responses:

(1) The PI and Intervention Coordinator reviewed the existing questions, comparing them to the rubric and making sure that they were worded appropriately. Our criteria for appropriate wording were:
    a. To the extent possible, questions should use the same wording as the sentence in the text that the question was about.
       *Example Text*: Life must be hard for the people who lived there.
       *Example Question*: Why must life be hard for the people who lived there?
    b. Questions should not include extraneous words that might confuse students or throw them off track.
       *Example Text*: The traveler pulled his serape, a colorful woolen shawl, over his shoulders.
       *Example Question*: Why did he pull his serape over his shoulders?
    c. If an important sentence could not be framed as a why question, the same criteria were applied to the creation of how questions.
       *Example Text*: I can find out whether there is anything I can do to help my people.
       *Example Question*: How can he find out whether there is anything he can do to help his people?
(2) Questions, along with follow-up questions (which could be used to clarify questions with ambiguous referents or that were overly long), and expected responses were entered into a table for each text.

(3) The questions and responses, along with the texts and the Question Development Rubric, were sent to co-PI van den Broek and Discourse Expert David Rapp. These two experts provided comments using track changes, including edits to existing questions or suggestions for different questions. In some cases, they suggested minor modifications to the texts.
(4) The expert feedback was used to make final question selections.

Once final questions were selected, they were inserted into the Question and Response Template (Appendix C).

## Results

Table 9 shows the Fountas & Pinnell (2006) benchmark passages, as well as texts selected for the intervention, and the readability and Text Easability Assessor dimensions of interest. Each text included 5-6 questions.

**Table 9.** ***Text Dimensions of Study Passages***

| Used For | Type | Text-Title | Words | F-K | N | SS | WC | RC | DC |
|---|---|---|---|---|---|---|---|---|---|
| Benchmark | Narrative | The New Girl (Level O) | 406 | 3.0 | 90% | 78% | 81% | 35% | 55% |
| Benchmark | Narrative | A Pet for Nate (Level P) | 409 | 4.4 | 57% | 80% | 94% | 3% | 27% |
| Benchmark | Narrative | Hiding Places (Level Q) | 494 | 4.0 | 57% | 91% | 91% | 10% | 80% |
| Benchmark | Narrative | The Election (Level R) | 530 | 4.4 | 78% | 54% | 68% | 11% | 51% |
| Spring Pilot | Narrative | Little Singing Frog Part 1 | 459 | 3.5 | 80% | 63% | 98% | 42% | 45% |
| Spring Pilot | Narrative | Little Singing Frog Part 2 | 584 | 4.4 | 75% | 48% | 98% | 36% | 49% |
| Spring Pilot | Narrative | A Different Aladdin Part 1 | 481 | 3.5 | 94% | 72% | 18% | 64% | 68% |
| Spring Pilot | Narrative | A Different Aladdin Part 2 | 452 | 3.2 | 92% | 76% | 36% | 27% | 60% |
| Spring Pilot | Narrative | A Different Aladdin Part 3 | 443 | 4.7 | 88% | 48% | 87% | 22% | 74% |
| Spring Pilot | Narrative | The Emperor and the Peasant Boy Part 1 | 356 | 3.9 | 45% | 84% | 90% | 19% | 35% |
| Spring Pilot | Narrative | The Emperor and the Peasant Boy Part 2 | 372 | 3.0 | 82% | 75% | 35% | 14% | 10% |
| Spring Pilot | Narrative | The Emperor and the Peasant Boy Part 3 | 406 | 3.7 | 73% | 82% | 82% | 31% | 55% |
| Post only | Narrative | The Spider on the Mountain | 235 | 3.6 | 80% | 74% | 99% | 34% | 45% |
| Pre-post | Narrative | Annie Helps a Dolphin | 233 | 3.6 | 85% | 86% | 74% | 36% | 41% |
| Pre-post | Narrative | Sam Saves a Squirrel | 242 | 3.0 | 81% | 82% | 92% | 34% | 37% |
| Pre-post | Narrative | Maria and the Lizard | 246 | 3.5 | 93% | 61% | 39% | 33% | 58% |
| Benchmark | Info | Snake Myths F & P (O) | 348 | 2.6 | 57% | 92% | 63% | 30% | 73% |
| Benchmark | Info | Animal Instincts F & P (P) | 351 | 4.7 | 50% | 88% | 88% | 44% | 37% |
| Benchmark | Info | Not Too Cold for a Polar Bear F & P (Q) | 393 | 4.8 | 44% | 82% | 94% | 78% | 22% |
| Benchmark | Info | Fishing Smarts F & P (R) | 504 | 4.4 | 48% | 89% | 92% | 51% | 84% |
| Spring Pilot | Info | Jane Goodall Part 1 | 370 | 4.8 | 50% | 77% | 92% | 41% | 38% |
| Spring Pilot | Info | Jane Goodall Part 2 | 360 | 2.8 | 93% | 77% | 33% | 48% | 31% |
| Spring Pilot | Info | Jane Goodall Part 3 | 342 | 4.6 | 64% | 93% | 90% | 17% | 87% |
| Spring Pilot | Info | Olympic Dreams: Part 1 | 371 | 5.3 | 80% | 90% | 68% | 32% | 81% |
| Spring Pilot | Info | Olympic Dreams: Part 2 | 375 | 6.5 | 47% | 77% | 81% | 40% | 25% |
| Spring Pilot | Info | Olympic Dreams: Part 3 | 319 | 6.0 | 62% | 93% | 81% | 20% | 64% |
| Spring Pilot | Info | Cesar Chavez - Part 1 | 292 | 4.5 | 55% | 99% | 76% | 29% | 47% |
| Spring Pilot | Info | Cesar Chavez - Part 2 | 294 | 3.8 | 57% | 98% | 90% | 29% | 57% |
| Spring Pilot | Info | Cesar Chavez - Part 3 | 287 | 5.5 | 54% | 99% | 86% | 67% | 98% |
| Spring Pilot | Info | Cesar Chavez - Part 4 | 306 | 3.7 | 44% | 99% | 87% | 37% | 84% |
| Post only | Info | Helen Keller | 230 | 4.6 | 85% | 79% | 61% | 95% | 74% |
| Pre-post | Info | Clara Barton | 227 | 4.4 | 68% | 93% | 98% | 79% | 99% |
| Pre-post | Info | Walt Disney | 242 | 4.4 | 62% | 90% | 94% | 32% | 94% |
| Pre-post | Info | Gutenberg | 243 | 3.9 | 44% | 96% | 83% | 53% | 100% |

Note. F-K = Flesch-Kincaid, N = narrativity, SS = syntactic simplicity, WC = word concreteness, RC = referential cohesion, DC = deep cohesion.

## DEVELOPING THE INTERVENTION

The purpose of developing the intervention was to identify one or more feasible and promising questioning approaches to improving struggling readers' comprehension of narrative and informational text. Our primary aims in Year 2 were to determine whether the intervention was similarly effective for different text types (stories vs. informational), as well as to find ways to individualize instruction in the context of a small-group intervention.

For the latter aim, we determined that one way to individualize instruction within the context of a small-group intervention would be to focus on the way we provide feedback to students' individual responses. This focus on feedback emerged because data from Year 1 indicated that feedback is a likely "active ingredient," in that students' connections improved when they were given feedback. We wondered whether the type of feedback given to students would matter. Thus, we decided to contrast two types of feedback: scaffolded and direct.

In the scaffolded feedback condition, if a student's response is only partially correct or incorrect, the tutor responds with follow-up questions designed to guide the student to making a good connection. If, after two attempts to scaffold the correct response, the student still does not make a good connection, the tutor models the correct response, by using the text to make a good connection. The direct feedback condition bypasses the scaffolding, and goes straight to a direct teacher modeling of the correct response. Pros and cons of scaffolded vs. direct feedback are listed in Table 10 below.

**Table 10.** ***Pros and Cons of Scaffolded versus Direct Feedback***

| SCAFFOLDED FEEDBACK | DIRECT FEEDBACK |
|---|---|
| Pros:<br>• Designed to be responsive to individual students<br>• Allows time for further text processing<br>Cons:<br>• May distract students from overall meaning of text<br>• May not be appropriate for all students in the group<br>• Time consuming; may decrease opportunities to read and respond to multiple questions | Pros:<br>• Easier to implement with fidelity<br>• Efficient<br>• May allow for more opportunities to read and respond to questions<br>• May be more engaging for whole group<br>Cons:<br>• May not respond to individual student's difficulty answering questions<br>• Does not necessarily allow for further text processing |

Specific research questions included:

(1) What is the fit and feasibility of the intervention for implementation in school settings?
(2) Does the effect of intervention vary based on text type (narrative stories vs. narrative informational) and feedback approach (direct vs. scaffolded)?
(3) Do subgroups of struggling readers (paraphrasers and elaborators) respond differentially to intervention?

Below, we first describe the 'final' version of the intervention to provide context for the description of its development. Then, we describe six 'Rounds' of iterative development and refinement that led to the current version of the intervention.

## Description of the Final Version of the Intervention: "In-the-Text Connections"

### Intervention Structure

The research staff (project coordinators, GRAs and URAs) served as tutors who implemented the instruction to participating students in small groups of 3 to 4, with the exception of one group that had five students. The intervention was conducted 3 times per week for 20 to 30 min per session in a small room such as the school's book room, media center, resource room, or in the hallway near the students' classroom.

The intervention was implemented in two Phases (in which students received the intervention using stories or informational texts in counterbalanced order). For the entire intervention period, 9 groups received Direct Feedback and 8 groups received Scaffolded Feedback. Each phase consisted of 9 instructional days, plus 1 introductory day (Day 1) at the beginning of Phase 1. Day 1 consisted of an introductory lesson in which students were taught the purpose of the activity, how to use materials, rules, and how points were awarded. The tutor also modeled how the students would be making connections in upcoming lessons. During Days 2 and 3, the groups read a text, and the tutor modeled how to answer the first question, followed by guided practice in which the tutor scaffolded steps needed to answer questions and provided structured feedback. From Day 4 onwards, students read and answered questions as a group. The tutor provided the students with direct or scaffolded feedback based on their answers. (More detail is provided below in the Protocol description).

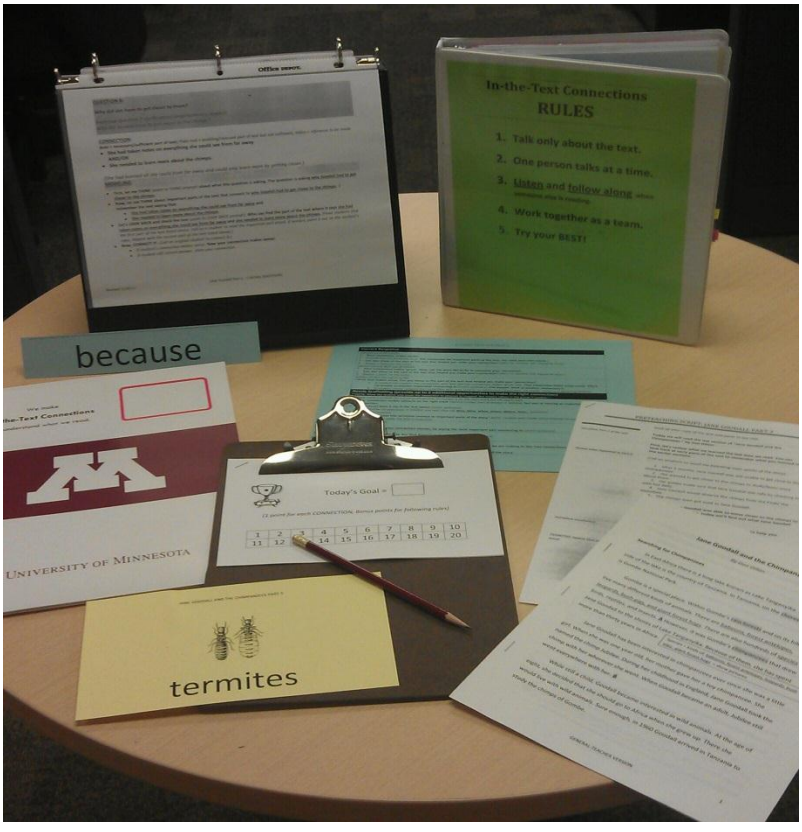
### Intervention Materials

Tutoring materials are displayed in Figure 3. Each student had a two-pocket folder with the purpose of the activity written on the front of the folder. Inside the two-pocket folder was a printed text that was divided into a separate part for each intervention session. The text was written in 14 point Calibri font on an 8.5 x 11" paper and stapled together into a booklet. At the end of sentences in which a causally-important connection could be made, a highlighted letter was inserted (see sample in Appendix D). The students also had a blue 8 x 1.5" bookmark with the word 'because' on it, to be used for making connections.

Tutor materials consisted of a two-pocket folder with a teacher copy of the text and a 'because' bookmark. The teacher copy of the text had underlined words with definitions for those words presented in text boxes in the page margin (see sample in Appendix E). The tutor could provide definitions to the group for the underlined words while reading, as needed. The tutor also had a three ring binder with 8.5 x 5" vocabulary cards and scripts for the preteaching component, teacher modeling and ongoing scripts, rules displayed on the front of the binder, and the Think-Look back-Connect it! prompt displayed on the back of the binder. There was also a presentation easel containing the following materials: rules; the Think- Look back-Connect it! prompt card, question cards that faced the student, and the teacher feedback cards that faced the teacher for each question. The question card had an answer stem to help students answer each question in a complete sentence. To aid the teacher with providing appropriate feedback, tutors were given a feedback prompt card with a modeling script. These feedback prompt and modeling cards were more general than the individual question teacher feedback cards in the binder and were tailored to the feedback conditions: direct and scaffolded (see Direct and Scaffolded feedback cards in Appendices F and G). There was also a point sheet on a clipboard. The tutor

would highlight the goal for the day (to indicate the number of points the group should try to receive) and kept track of points by marking a slash through a number each time a point was earned.

**Figure 3.** ***Sample Tutoring Materials for Final Intervention Version***




**Intervention Protocol**

A typical intervention session (following Introductory and training Days 1-3) consisted of two parts: (1) preteaching, and (2) guided practice with feedback. Each intervention session started with goal setting which involved establishing the number of points the group would strive to earn and writing this number on the point sheet.

**Preteaching**. The preteaching component followed goal setting and was conducted for no more than 5 min. Preteaching involved activating background knowledge and teaching definitions of two to four words that were central to the text. Activating background knowledge consisted of providing students with a brief preview of the text, asking students questions on similar texts they may have read, and/or asking students to discuss related experiences they may have had.

The vocabulary words were highlighted on the student and teacher copies of the text. When introducing the vocabulary words, the tutor first showed the word printed on a card and said the word aloud. Next, the student read the word aloud with the tutor and then pointed to the same word highlighted in the text. Then, the tutor provided a definition for the vocabulary word and/or a picture to depict the meaning of the word. The students repeated the definition back to the tutor. Next, the tutor asked a question pertaining to the word and the student had to respond to the question using the word. This procedure was used to give students several opportunities to

actively engage with the vocabulary word. The preteaching component procedure was the same for both feedback conditions.

**Guided practice with feedback**. The remainder of the session was devoted to reading and making connections. First, the tutor read the text aloud up to the first highlighted letter. The tutor provided meanings to unknown words while reading, as needed. Next, the tutor selected a student to reread the same section. If time was limited, the reading was only done once either by a student or the tutor. Upon reaching the letter, the tutor asked a causal question, which was displayed on the presentation easel, and prompted students to think of a connection, look back to check their connection in the text, and share their connection with a peer. After students their connections with each other, the tutor called on one student to 'Connect it,' and the student shared his/her answer.

The tutor gave structured feedback based on the student's answer. The feedback provided was dependent on the condition of the group, either Direct or Scaffolded. In the Direct feedback condition, when a student's connection was not correct, tutors provided an immediate, explicit model of how to make the connection. In the Scaffolded feedback condition, when a student's connection was not correct, tutors attempted to guide them through processing the text. If the connection was still incorrect after two prompts, the tutor provided an explicit model of how to make the connection just as in the direct feedback condition.

The group received points for making connections and for working together and following the rules. The tutor continued the process of reading and answering questions until all questions for the day were answered or till the last 3 minutes of the session. At the end of each session, if the group met their goal, students received stickers to put on their folders.

Below, we describe the iterative development cycle that led to the version of the intervention described above. We begin with "Round 6," picking up from Rounds 1-5 conducted in Year 1.

### Round 6: In-House Development (August-November, 2012)

Project staff (PI, project coordinators, GRAs) reviewed the Days 1-3 scripts in detail, discussing aspects that were troublesome during the Year 1 trials and pilot study.
Scripts were revised based on these discussions and data sources from Year 1, including student and tutor survey feedback on the text, tutor logs, and the text review questionnaire completed by reading specialists. The student survey feedback was used to determine student preference for the texts used. The tutor survey feedback was used to determine age appropriateness, interest, and level of difficulty for each text. The tutor feedback on various components of the intervention such as modeling, preteaching, partner sharing, and prompting given to help student make a connection was used to determine changes to be made for Year 2. Tutor logs were also reviewed to examine which questions in individual stories needed to be modified, including the answers/connections to the questions. An experienced GRA delivered two "mock" intervention sessions to volunteer children in a video-recording session (to be used to gain feedback from intervention experts and teachers). Additional tweaks were made to the intervention based on trouble spots that emerged during the video recording.

## Round 7: Focus Groups (November, 2012) and Expert Feedback (December, 2012)

The purpose of Round 7 was to obtain feedback from (a) teachers on the intervention's fit with their schools' instructional programs and students' needs and feasibility to be implemented within the schools' instructional structures and schedules, and (b) experts in the field on the intervention's viability and likelihood to produce desirable outcomes.

### Participants

All participating teachers were invited to attend one of three half-day focus group sessions. In Session 1, 8 classroom teachers and two specialists attended. In Session 2, 11 classroom teachers and 5 specialists attended. In Session 3, 11 classroom teachers and 5 specialists attended. One 'make-up' session was held for three classroom teachers who had been unable to attend any of the other 3 sessions.

### Measures

The Focus Group Questionnaire and Intervention Expert Feedback Form, described under procedures, are in Appendices H and I.

### Procedures

All focus group sessions followed a similar format. First, project staff provided an overview and rationale for the intervention, as well as a timeline of project activities and description of how students were identified as eligible. In addition, Year 1 results were briefly presented, with opportunities for discussion and questions regarding Year 1 findings. Then, the teachers viewed segments of the video, and followed along on 'student' copies of the story used during the video. After watching the video, the teachers completed the Intervention Questionnaire, which asked them to rate various components of the intervention based on the Fit and the Feasibility of implementation in a school-based setting.

The core intervention components rated were interventionist, group size, session duration, frequency, text used, preteaching component, questioning approach, number of questions, and response mode. Next, the group discussed the various components of the questionnaire, and asked specific questions about the intervention after watching the video. The questionnaire was used to get feedback and suggestions on how to improve the fit and feasibility of the intervention in a school-based setting. The teachers also filled out a logistics questionnaire to help the research staff develop a timeline for implementation of the Spring Pilot.

Three intervention experts gave feedback on the intervention and completed the Intervention Expert Feedback Form. The experts first watched the video, and then answered open-ended questions giving their general reactions, questions, comments, ideas, and concerns about the intervention. Next, the experts completed open-ended questions on specific components of the intervention such as preteaching, modeling, and guided practice with feedback. They also gave feedback on whether the intervention has potential to help struggling readers and suggestions for improvement, and the components that would be essential to include in teacher training during intervention delivery. Finally, the experts gave feedback on the research design for Year 2 of the study.

**Round 7 Revisions**

**Focus group feedback**. Table 11 shows the categories of feedback and examples of revisions made based on focus group feedback.

**Table 11.** ***Focus Group Feedback Revisions***

| Teacher Input or Suggestion | Revision to the Intervention |
|---|---|
| Clarify what kind of connection, avoid confusing students who have understood other connections taught in their classes | Use new terminology: In-the-Text Connection; compare to text-to-text, text-to-self, text-to-world. |
| Reading options: Silent reading before oral reading? Repeat reading? Repeat reading by students or by tutor? | Repeated reading was incorporated: tutor read first followed by student reading. In some groups, two students read aloud. |
| Need explicit modeling of what we mean by a connection | Teacher models during the first two days of instruction; both feedback conditions include modeling when students need help. |
| Connection-making should be more explicit with kinesthetic or visual representation | Have students point to the text/sentence that makes the connection. |
| Think-Look Back-Connect It should be maintained as three steps so students develop the routine | Tutors reinforced all three steps (Think-Look Back-Connect It). Three steps are faded during a release of responsibility process. |
| Avoid judgmental language: "good connection" | Provide neutral feedback: "That connection makes sense!" |
| Partner sharing: if partner shares partner's connection give a chance to correct; allow partner who did not share to add to the connection | Tested variations during trial run. Partner sharing became collaborative. One partner shares his/her connection, but can incorporate partner ideas. |
| Reinforce team effort by letting student ask for help from other students; let first student complete the connection | Teamwork is reinforced. Student who first attempts a connection is given another chance to complete the connection after help from other students or teacher modeling. Changed rule from "help each other" to "work together as a team." |
| Vocabulary preteaching is good; keep cards visible during reading; include pictures | Vocabulary cards include pictures whenever possible, and are kept visible on table during reading group. |
| Gradual release of responsibility is a good goal | Release of responsibility tested during the May trial run. |

**Expert feedback**. Based on feedback from our two experts (Deno and Jenkins), we made the following revisions: (1) Made the feedback more flexible in order to accommodate students' connections that made sense, but were not necessarily anticipated or aligned with what we deemed to be 'correct;' (2) Continued to scale back the teacher talk; (3) Clarified teacher wait time between asking a question and prompting students to make connections.

## Round 8: Fall Trial Run (December, 2012)

The purpose of the trial run was to determine the feasibility and flow of the structure of the intervention. In other words, we examined whether the intervention was operating as intended. We were particularly interested in whether revisions made up through Round 8 were operating as intended (e.g., the scripts were clear and flowed well). We also wanted to test several different approaches to reading, including round robin, repeated reading, partner reading, and silent reading, and to determine the best approach to getting students to successfully share connections with a peer partner.

### Participants

The trial run was conducted with a subset of students who just missed meeting the eligibility criteria for the intervention (n = 12). The 'just missed' eligibility criterion was a CBM maze score in the 14.5 – 15.5 range. Students were placed into 3 groups of 4 students based on their school schedules.

### Measures

**Intervention logs**. Tutors (two experienced GRAs) implemented the Trial Run instruction. They maintained Intervention Logs of each session that included the total number of minutes of instruction, student attendance, and the number of questions answered. On Day 1, tutors responded to 12 prompts using a 4-point Likert-type scale assessing their agreement (1=disagree to 4=agree) on topics covering perceived student understanding, student motivation, and materials. On the remaining days, tutors responded to 8 prompts and the corresponding 4-point Likert-scale to assess the utility of that day's reading approach, appropriateness of the text and questions, and perceived student motivation. Tutors were also asked to record which reading approach they would use on the following day and explain their rationale for their selection. For Day 2 onward, tutors were asked to note specific issues/problems and suggestions for improvement for items rated lower than 4. Tutors brought their intervention logs to lab meetings to facilitate discussions regarding whether the intervention was operating as intended or additional revisions were needed. Mean ratings were obtained for each component, and open-ended feedback was summarized and used for further revision to the intervention. The Fall Trial Run Intervention Log is found in Appendix J.

**Fidelity draft**. A draft of the fidelity form revised from Year 1 was used to observe the Trial Run. This form is described in detail under Round 9 below, and was further revised between the Fall Trial Run and Spring Pilot Study.

### Procedures

The trial run was conducted for two weeks in December. During the trial run, each member of the research team who was not serving as a tutor (PI, project coordinators, 3 GRAs) observed each tutor at least once, using the Fidelity draft.

At the end of each week, all research staff met to discuss the intervention, using the Intervention Logs and Fidelity drafts to guide discussion. Detailed notes were taken during these meetings, and Intervention Log data, including ratings and comments, were summarized. Each category of feedback was reviewed systematically, discussed among the PI, co-PIs, and project coordinators during Skype meetings, and incorporated into revisions of the intervention scripts and materials. Revisions were reviewed by co-PIs van den Broek and Espin.

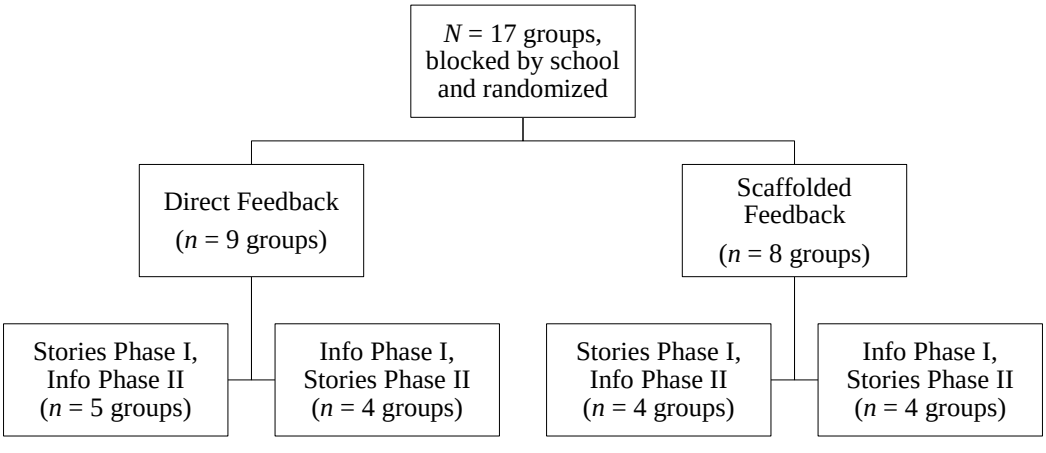
**Round 8 Revisions**

The most significant revisions made based on the Trial Run were (a) revising the Day 1 introductory script to make it less disjointed, (b) clarifying the purpose of the intervention, (c) making general proofreading edits, (d) adding significant amounts of practice to tutor training sessions for the Spring Pilot.

## Round 9: Spring Pilot (Late January through late March 2013)

The purpose of the Spring Pilot was to obtain preliminary data regarding the extent to which the revised Causal questioning intervention showed promise for subgroups of struggling comprehenders, and whether students' responsiveness to the intervention varied with respect to text type (stories vs. informational) and type of feedback (direct vs. scaffolded). We used a quasi-experimental, randomized mixed design with one between-groups (FEEDBACK: direct vs. scaffolded), one within-subjects (TEXT: stories vs. informational) variables, as illustrated in Figure 4.

**Figure 4.** ***Assignment to Feedback Condition and Text Type***




**Participants and Group Assignment**

All students identified as eligible for intervention participated (n = 61). Because the intervention was conceived as a small-group intervention, students were placed in groups of 3-4 (and one group of 5) within their schools (within or across classrooms, as scheduling dictated). Groups (n = 17) were assigned randomly to Direct vs. Scaffolded feedback conditions within school (when possible) or within two schools with similar demographic make-ups (when within school was not possible). Direct and Scaffolded groups were then assigned randomly to either receive the narrative or narrative information texts in Phase I (Weeks 1-3), and the other text type in Phase II (Weeks 4-6). Demographic and screening data for students in the two groups are provided in Table 12, along with chi-square and *F*-statistics. There were no statistically significant differences between the two groups in terms of demographic or screening (pre-treatment) performance.

**Table 12.** ***Student Demographics***

| | Total ($n$ = 61) | | Direct ($n$ = 32) | | Scaffolded ($n$ = 29) | | | |
|---|---|---|---|---|---|---|---|---|
| | *n* | % | *n* | % | *n* | % | $X^2$ | *P* |
| Sex (M) | 29 | 47.54% | 13 | 40.63% | 16 | 55.17% | 1.29 | .256 |
| ELL | 18 | 29.51% | 8 | 25% | 10 | 34.48% | .66 | .417 |
| Free/Reduced Lunch | 44 | 72.13% | 23 | 71.88% | 21 | 72.43.38% | 0.002 | .963 |
| Race | | | | | | | 2.02 | .846 |
| African American | 20 | 32.79% | 10 | 31.25% | 10 | 34.48% | | |
| Asian | 6 | 9.83% | 4 | 12.5% | 2 | 6.90% | | |
| Hispanic | 16 | 26.23% | 7 | 21.88% | 9 | 31.03% | | |
| White | 16 | 26.23% | 9 | 28.13% | 7 | 24.14% | | |
| Other | 3 | 4.92% | 2 | 6.25% | 1 | 3.45% | | |
| Special Education | 9 | 14.75% | 7 | 21.88% | 2 | 6.90% | 2.71 | .099 |
| | Mean | *SD* | Mean | *SD* | Mean | *SD* | *F*(1, 60) | *P* |
| Age (end of year) | 10.19 | .27 | 10.19 | .28 | 10.19 | .28 | .004 | .953 |

Tutors were assigned to groups in such a way as to minimize any tutoring effect between the Direct and Scaffolded groups. Specifically, we attempted to ensure that either each tutor worked with one group in each condition, or that people with similar levels of experience were represented in both conditions. Table 13 shows how tutors with different levels of experience were distributed across groups.

**Table 13.** ***Assignment of Tutors to Groups***

| Direct Feedback | Scaffolded Feedback |
|---|---|
| Tutor 1 (Project Coordinator; highly experienced with children, worked in both Years 1 & 2 of project) | Tutor 2 (Project Coordinator; highly experienced with children, worked in both Years 1 & 2 of project) |
| Tutor 3 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) | Tutor 3 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) |
| Tutor 4 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) | Tutor 4 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) |
| Tutor 5 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) | Tutor 5 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) |
| Tutor 6 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) | Tutor 6 (GRA; highly experienced with children, worked in both Years 1 & 2 of project) |
| Tutor 7 (GRA; highly experienced with children, worked only in Year 2 of project) | Tutor 7 (GRA; highly experienced with children, worked only in Year 2 of project) |
| Tutor 8 (GRA; highly experienced with children, worked only in Year 2 of project) | Tutor 8 (GRA; highly experienced with children, worked only in Year 2 of project) |
| Tutor 9 (URA; some experience with children, worked only in Year 2 of project) | Tutor 9 (URA; some experience with children, worked only in Year 2 of project) |
| Tutor 10 (URA; some experience with children, worked only in Year 2 of project) | |

**Measures**

Several types of measures were used during the Spring Pilot, including measures of fit and feasibility, fidelity, social validity and promise of the intervention.

**Fit and feasibility**. Fit and feasibility feedback was solicited from both tutors who implemented the intervention and from reading specialists who observed it. Tutors maintained logs of each session that included the total number of minutes of instruction, student attendance, and the number of questions answered. These data were compiled for both group and individual dosage information (total sessions completed, total minutes completed, and total questions answered). Space was also provided for tutors to note any special issues related to individual student attention and behavior, as well as an special issues related to the session such as what went well, didn't go well, and suggestions for improvement. A Spring Pilot Intervention Log is in Appendix K.

Also during the Spring Pilot, four of the participating specialists observed the intervention and provided feedback on an Intervention Questionnaire. This questionnaire asked the specialists to 'rate each core intervention component on its fit with instructional programming and student needs in your school, as well as its feasibility for implementation in school-based settings' and to elaborate on their responses with comments and suggestions. Items to be rated included the interventionist (adult tutor), group size, duration and frequency, text type, pre-teaching activities, reading approach (repeated reading with adult model), questioning approach, number of questions, response mode, feedback & modeling, scripted nature of the intervention. Items were rated on a 1-4 Likert-type scale, and included both fit (1 = not a good fit; 4 = very good fit) and feasibility (1 = not feasible; 4 = highly feasible). A final item asked the specialists to compare student engagement during the intervention to student engagement during other small-group reading interventions. This item was rated on a five-point scale ranging from much less engaged (1) to much more engaged (5). Reading specialists were also given space to respond to an open-ended question, which asked for any other comments, suggestions or questions about the intervention. The Intervention Questionnaire is provided in Appendix L.

**Fidelity**. The fidelity observation form developed in Year 1 study was revised to include the Scaffolding and Direct feedback approaches used in Year 2. The 'Scaffolded' section was highlighted, so that observers were alerted to count as positive scaffolding behaviors that were correctly implemented in the Scaffolded feedback condition, and to count as negative scaffolding behaviors that were implemented during the Direct feedback condition. The form included several items under each of the following sections for Set-up (3 items); Introduction (3 items); Pre-teaching (4 items); Delivery, Motivation, and Engagement (4 items); Ongoing Instruction (8 items); and Teacher provides feedback (6 items for Direct feedback and 9 items for Scaffolded feedback). Some items were rated as 0 or 1 (e.g., 'student materials are ready'), whereas others were rated on a scale of 0 to 3 (0 = No, 1 = Somewhat, 2 = Mostly, 3 = All/Yes; e.g., 'Teacher is familiar with script;' 'Students appear to be engaged for most of the session'). Also, because each story had up to six questions (labeled A-F), items under Ongoing Instruction were rated for each question (e.g., for 'Teacher prompts students to think, look back, connect it,' the rater would circle the letter for each question for which this behavior was observed). The final version of the Fidelity Form is in Appendix M.

**Student feedback**. Student feedback was solicited from students who participated in the intervention. Students completed a questionnaire, which was read aloud to them individually

during posttesting at the end of the Spring Pilot. Questions asked students to rate the helpfulness of components of the intervention (e.g., 'How helpful was it to answer questions and make connections to understand the stories we read during Making the Right Connections?') and how much they liked components of the intervention (e.g., 'How did you feel about setting a goal for the number of points to earn each session). Students also rated how well they liked each of the stories they read during the intervention. All items were rated on a 1-4 Likert-type scale (1 = very unhelpful to 4 = very helpful, or 1 = I disliked a lot to 4 = I liked a lot). Thumbs-up/thumbs down pictures were paired with the ratings to make sure it was clear what the student was being asked to do. Following the ratings, students were asked three open-ended questions: (1) 'Which type of text did you prefer reading?' (multiple-choice response: stories, biographies or none of the above) (2) 'When you are reading a story, what are some things you can do to help you understand what you are reading?' (3) 'Would you recommend the Making the Right Connections activity to other kids like you? Why or why not?' The student questionnaire is in Appendix N.

**Tutor feedback**. Tutors also completed a questionnaire after the Spring Pilot was completed. This questionnaire covered a wide range of topics covering all aspects of the intervention. The questionnaire began by asking a mix of multiple choice, Likert-type rating scale and open-ended questions about intervention preparation and overall perceptions of the intervention. This was followed by 20 Likert-type and multiple choice questions on specific details of the intervention implementation. The remainder of the questions concerned the tutor's overall perception of the intervention. These questions asked the tutor to reflect on their perceptions of various aspects of the intervention. This section included an open-ended question, which asked tutors to discuss any changes they would make, and how these changes could be implemented. The tutor questionnaire is in Appendix O.

**Promise**. The Spring Pilot study was conducted in two Phases (as described in Procedures). After each phase, posttests were given. Below, we describe posttest texts and dependent variables.

**Pre- and posttest texts**. During the pretest, students read one of three pairs of 'far-transfer' texts comprised of one narrative and one informational text. During Phase I and Phase II posttesting, one of two 'near transfer' texts that was either narrative or informational based on which type of text the students had been reading during the current intervention phase. One of three pairs of 'far transfer' texts that was not used in the pretest was used at each posttest. The texts were similar in format and readability level to those read during the intervention. The order of the 'far transfer' texts were counterbalanced across pretest and Phase I and II posttests to account for order effect.

Near transfer texts were administered in a similar format to the format used during training (with the same types and timings of questions asked), and far transfer texts were administered without the questioning. Including near- and far-transfer texts allowed us to examine immediate effects of the questioning intervention, as well as whether intervention effects transferred when the intervention was no longer provided. The near-transfer informational text was "Helen Keller" and the narrative was "The Spider on the Mountain." Both near-transfer texts were designed to be parallel to the informational and narrative far transfer texts in the same way as the far transfer texts were related. The pairs of far transfer texts included one narrative and one informational text. The three narrative far-transfer texts, "Annie Helps a Dolphin," "Maria and the Lizard" and "Sam Saves a Squirrel," were designed to be parallel to each other,

in that they followed a very similar story structure and general plot with similar goals, sub-goals, and resolutions, but with different characters and specific scenarios. The three informational texts "Clara Barton," Johannes Gutenberg" and "Walt Disney" were used due to their similarities in length, readability, and biographical structure. "Annie Helps a Dolphin" was paired with "Walt Disney" (Pair A), "Maria and the Lizard" was paired with "Johannes Gutenberg" (Pair B) and "Sam Saves a Squirrel" was paired with "Clara Barton" (Pair C). All texts were similar to each other and to stories used in the intervention in terms of readability and the five Text Easability features. Pairs of texts were counterbalanced such that half of the students in each condition (Question Type and Timing) were administered different pairs at pretest, posttest Phase I, and posttest Phase II. Table 14 shows the counterbalanced order of text pairs administered by condition in Phases 1 and 2. Administration instructions are in Appendices P-Q. Examiner versions of the posttest protocols are in Appendices R-U.

**Table 14.** ***Counterbalancing of Posttest Text Pairs by Condition and Phase***

| | | | | Phase 1 | Phase 2 |
|---|---|---|---|---|---|
| | Phase 1 | Phase 2 | Pretest | Posttest | Posttest |
| Feedback Type | Text Type | Text Type | Text | Text | Text |
| Direct | Story | Informational | Pair A | Pair B | Pair C |
| | | | Pair A | Pair C | Pair B |
| | | | Pair B | Pair A | Pair C |
| | | | Pair B | Pair C | Pair A |
| | | | Pair C | Pair A | Pair B |
| | | | Pair C | Pair B | Pair A |
| | Informational | Story | Pair A | Pair B | Pair C |
| | | | Pair A | Pair C | Pair B |
| | | | Pair B | Pair A | Pair C |
| | | | Pair B | Pair C | Pair A |
| | | | Pair C | Pair A | Pair B |
| | | | Pair C | Pair B | Pair A |
| Scaffolded | Story | Informational | Pair A | Pair B | Pair C |
| | | | Pair A | Pair C | Pair B |
| | | | Pair B | Pair A | Pair C |
| | | | Pair B | Pair C | Pair A |
| | | | Pair C | Pair A | Pair B |
| | | | Pair C | Pair B | Pair A |
| | Informational | Story | Pair A | Pair B | Pair C |
| | | | Pair A | Pair C | Pair B |
| | | | Pair B | Pair A | Pair C |
| | | | Pair B | Pair C | Pair A |
| | | | Pair C | Pair A | Pair B |
| | | | Pair C | Pair B | Pair A |

Note. Pair A = Annie Helps a Dolphin & Walt Disney; Pair B = Maria and the Lizard & Johannes Gutenberg; Pair C = Sam Saves a Squirrel & Clara Barton; see Pretest and Posttest section for description of how the order of the texts within the pairs and phases were determined.

**Dependent variables**. The following dependent variables were measured using near and far transfer texts:

***Responses to questions using intervention approach*** (near transfer only). The student read the post-intervention text aloud, responding to 5 causal online questions with connection-making feedback that matched the scaffolded or direct feedback delivered during the intervention. The score was the number of prompts required and whether a correct connection was made with or without prompting. See Appendices V-X for questions and feedback.

***Verbal recalls*** (near and far transfer). After reading and responding to questions (for the near transfer text) and after reading only (for the far transfer text), the student was asked to recall the text just read. Specifically, the students were prompted, "Now, let's pretend that I am a friend from your class, and I want to know all about what happened in the text. Please tell me as much of the text as you can remember. What happened first?" The student was additionally prompted, "What happened next?" until the student finished retelling. If the student did not tell the very end of the text, the student was finally prompted, "Is there anything else about the text that you'd like to tell me?"

Students' responses during the recall of each text were audio recorded, transcribed, and parsed into thought units based on the original text. Each recalled thought unit was compared to the parsed story units in the original text and coded using a scheme adapted from Kendeou and van den Broek (2005). The coding included the original story unit number to which it matched (if any), and the extent to which it matched or was constrained by the text. The following five codes were recorded:

- Conservative (C; directly matches a story unit and is a literal or nearly literal rendering of the original story unit)
- Liberal (L; directly matches a story unit and captures the gist of the story unit (the essence of its meaning; paraphrasing may be present)
- Matched but incorrect (MI; directly matches a story unit, but is significantly incorrect. MI is reserved for absolutely and importantly contradictory interpretations, not just an erroneous detail of the story)
- No match consistent (NMC; cannot be matched directly with the gist meaning of any one or two adjacent story units, but is valid, consistent with/constrained by the information in the text, or plausible.)
- No match inconsistent (NMI; invalid, inconsistent (unconstrained by) the information in the text. The thought unit should be flat-out incorrect to be NMI)

The number of matches to highly connected original story units was also counted. Highly connected story units were those that had at least six or seven (depending on the text) connections to other story units, or stated a goal or attainment of a goal in the story. The recalls were also coded for order of recall, which was the correlation of the sequence of recalled thought units with sequence of the original text story units.

It should be noted that the matched but incorrect code was new in Year 1. The research team believed it was useful to distinguish between recalls that accurately reflected the story unit (conservative or liberal) from those that matched a story unit but reflected a significant misunderstanding. The matched but incorrect code was applied cautiously for thought units that directly contradicted the story unit in way that would disrupt coherence.

***Words read correctly in one min*** (near and far transfer). After completing the verbal recall of the near or far transfer text, students re-read the text aloud for one min, following the same procedures used during the CBM oral reading task described above. WRC were recorded.

**Procedures**

**Tutor training**. The two project coordinators, seven GRAs and three URAs served as tutors during the Spring Pilot. Before intervention began, tutors participated in two four-hour training sessions. During the first session, we (a) reviewed the purpose of the study, research questions, timeline, and design; (b) reviewed in detail and demonstrated the intervention procedures and materials; (c) discussed key considerations for implementation, such as delivering all components with fidelity, ensuring multiple opportunities to respond, maintaining student engagement, providing appropriate modeling and feedback, and managing behavior; (d) reviewed tutor assignments to schools and groups; and (e) provided extended time for practice.

During the second session, we (a) went over logistics of delivering the intervention and maintaining tutoring logs, (b) provided pre-testing training, (c) demonstrated the Day 1 introductory lesson, (d) modeled and provided extended opportunities to practice ongoing instruction. Following training, tutors were instructed to continue practicing with a partner. When they were comfortable with the intervention, they met with the PI or a project coordinator to 'check out'—they needed to be able to implement all components of the intervention and use the materials accurately (at least 90% based on the revised fidelity tool) and fluently (e.g., making eye-contact, using a conversational tone of voice, not over-relying on scripts).

Following training, the research team met weekly for two hours. The first half of the meeting was devoted to problem solving: tutors raised issues they encountered during their tutoring sessions, such as behavior problems, maintaining student engagement, helping students understand the prompts, etc. The group discussed and generated ideas for addressing these issues. The second half of the meeting was devoted to previewing the texts and practicing the intervention for the upcoming week. Tutors met in 'text groups' – based on whether they were implementing the intervention with stories or informational texts. They read the text for the upcoming week, and made sure the questions and possible responses were clear.

**Intervention implementation**. Implementation began in late January and lasted through the end of March. All students were pretested. Then, each tutor began Phase I with a generic introductory lesson, followed by 9 lessons using their group's assigned feedback condition (Direct or Scaffolded) and text type (Story or Informational). Phase I was followed by individual posttesting sessions, which took approximately one week (no intervention was implemented at this time). When posttesting was complete, the tutor implemented Phase II, which consisted of 9 lessons. Phase II lessons were delivered using the same feedback condition as in Phase I, but the opposite text type.

**Fidelity observations**. During the Spring Pilot, two researchers observed 4 out of 17 intervention groups (24%) together during Phase 1 to obtain interrater agreement. Interrater agreement, calculated as the number of agreements divided by the number of agreements plus disagreements, ranged from 88.24% to 98.1% (mean = 95.75%). After establishing interrater agreement, the two researchers observed each tutor at least once during Phase I and again during Phase II. Another IRA check during Phase II indicated 98.59% agreement.

**Recall coding training and reliability**. Recall coding guidelines from previous studies were revised for the current study. During training and practice, the team revised and refined glossaries of exemplary coded thought units for each text, and co-PIs van den Broek and Espin parsed and conducted causal analyses of new texts (one new story and 4 informational texts). See Appendix Y for recall coding procedures.

The glossaries and the procedure document were the basis of GRA training on recall coding. Conducting training of GRA coders and practicing by the whole team took approximately two weeks. The team discussed and practiced applying the coding procedure as a group. Then, groups of GRAs practiced coding three protocols jointly with a project coordinator. Project coordinators also practiced coding as pairs. Questions were discussed and resolved at team meetings by the PI or by the coding experts; examples were added to glossaries. Coding pairs then coded five protocols jointly and the PI performed IRA for each coding pair. When pairs achieved at least 75% IRA on codes and on story unit numbers, the GRAs in the pair were approved to code independently.

After coding reliability was demonstrated, pairs coded one protocol in common for every two protocols coded independently. The PI made IRA checks on independent and consensus-coded protocols. Ten percent of protocols for each pair were selected randomly for IRA. Random selection of 10% of each set of protocols assigned to individuals or pairs was repeated for IRA until all coding was complete.

The IRAs on recall coding were calculated for agreement on codes and agreement on story unit numbers of conservative or liberal matches. Note that until the third party IRA with an individual or a pair was at least 75%, PI coding and IRA checking continued. Coding is still in process, so final IRA percentages are not yet available.

During the two months of coding, the PI, PCs, and GRAs met weekly to practice coding of at least one recall protocol to maintain calibration. The protocols that had been practice coded during the training period were recoded in the routine manner toward the end of the coding period.

**Data coding and entry**. After both phases of intervention were complete, all data were transcribed (as needed), coded, and entered into an excel spreadsheet. All relevant codes were counted, and counts were double-checked by a second coder. All data were entered by pairs of GRAs, who entered all data twice into two separate excel worksheets. For the first entry, one GRA called out the data and the other entered the data. For the second entry, the GRAs switched roles. The two worksheets were then compared using the logical test formula in Microsoft excel, and all discrepancies were resolved. Excel sheets were converted to SPSS 21.0 for analysis.

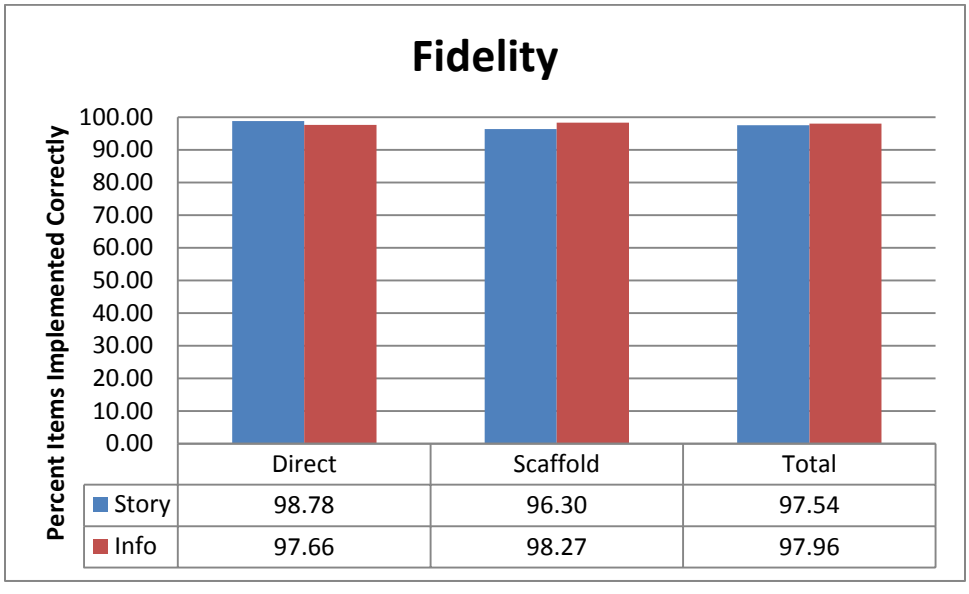
## Results

**Research Question 1:**
What is the fit and feasibility of the intervention for implementation in school settings?

**Fidelity**. Fidelity observations were conducted to determine whether the tutors implemented the intervention as intended. Whereas this information is important for interpreting student response to the intervention, it is also important to know whether the tutors were able to implement the intervention based on the training and support that they received to implement it. Figure 5 shows the mean proportion of fidelity with which the intervention was implemented across text type (stories vs. informational) and feedback type (Direct vs. Scaffolded) conditions.

Generally, tutors implemented all versions of the intervention with high levels of fidelity, suggesting it was feasible to implement accurately.

**Figure 5.** ***Fidelity Percentages by Type of Feedback***




**Tutor ratings**. Tutors completed ratings of the intervention components, and provided open-ended feedback to elaborate on their ratings. Table 15 provides a summary of their ratings (all ratings were on a 1-4 scale, with 4 being more positive). Overall, tutors provided positive ratings. Open-ended feedback will be used to inform further revisions to the intervention.

**Table 15.** ***Tutor Intervention Ratings***

| Question | Rating |
|---|---|
| How helpful was answering questions to understand the stories? | 3.4 |
| How helpful was answering questions to understand biographies? | 3.5 |
| How helpful was making in-the-text-connections to understand the text? | 3.6 |
| How helpful was it to "Think of a connection, Look back, and Connect it"? | 2.9 |
| How useful was the direct feedback? | 2.9 |
| How useful was the scaffolded feedback? | 4.0 |
| How helpful was the preteaching to activate background knowledge? | 3.6 |
| How helpful was partner sharing to maintain student engagement? | 3.4 |
| How helpful was setting a goal? | 3.6 |
| How helpful was giving points? | 3.6 |
| How helpful was giving stickers when the goal was met? | 3.9 |
| How appropriate was it to answer 5 questions per session? | 3.3 |
| Was 30 minutes per session enough of time to implement the intervention? | 3.4 |
| Was 3 times per week sufficient to implement the intervention? | 3.4 |
| Was 3 to 4 students per group appropriate to implement the intervention? | 3.5 |
| Was the script helpful in leading the intervention? | 3.6 |
| Asking “Why” questions helps students find important connections in the text. | 3.4 |
| Asking “How” questions helps students find important connections in the text. | 2.9 |

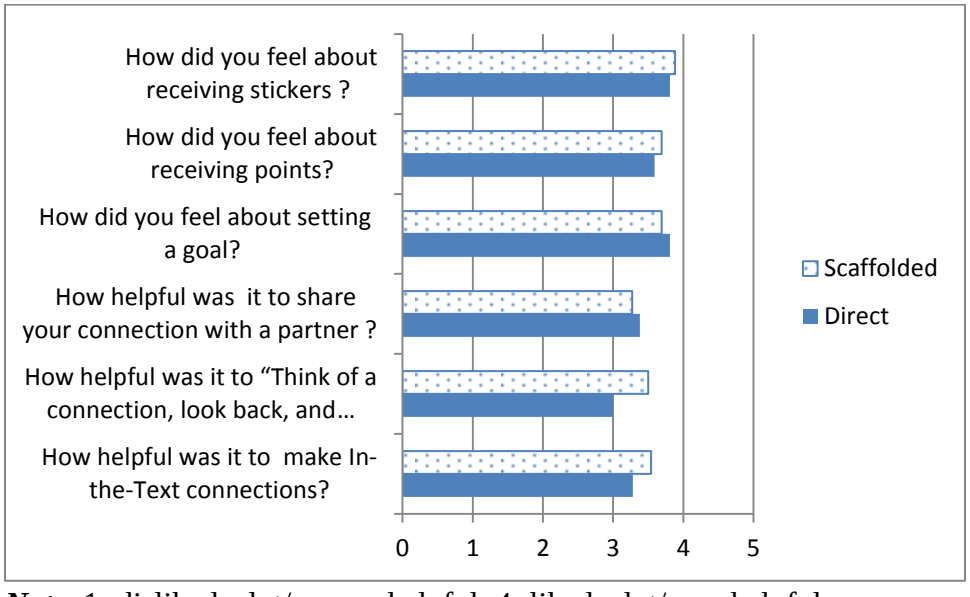
**Specialist ratings**. Four specialists observed implementation of the intervention, and completed ratings of the fit and feasibility of the intervention components. They also provided open-ended feedback to elaborate on their observations. Table 16 provides a summary of their ratings. Overall, specialists rated the intervention as having a good to a very good fit, and good to very good feasibility. They particularly liked the pre-teaching component.

**Table 16.** ***Specialists' Ratings of Intervention Fit and Feasibility***

| Category | Average Fit | Average Feasibility |
|---|---|---|
| **Interventionist: Adult tutor** | 3.80 | 3.00 |
| **Group size: 4 students** | 3.80 | 4.00 |
| **Session Duration: 30 min** | 3.80 | 3.20 |
| **Frequency: 3 times per week** | 3.60 | 3.60 |
| **Text: Narrative fiction and non-fiction (implemented in parts across 2-4 sessions)** | 3.80 | 3.60 |
| **Pre-teaching (designed to preview key vocab/concepts needed to understand the story)** | 4.00 | 4.00 |
| **Reading Approach (repeated reading with a model)** | 3.60 | 3.75 |
| **Questioning Approach (designed to help students make connections within text)** | 3.20 | 3.80 |
| **Number of Questions: Approximately 6 per text** | 4.00 | 3.80 |
| **Response mode: Partner share followed by individual responses** | 3.60 | 3.80 |
| **Feedback & Modeling (Provided as needed to help students make in-the-text connections that make sense)** | 4.00 | 3.80 |
| **Scripted nature of the intervention (designed to ensure fidelity to the intervention protocol)** | 3.60 | 3.80 |
| **Student engagement: Compared to other small-group interventions, how would you rate the engagement of students in this intervention**<br><br>(Ratings: 1 = much less engaged, 3 = similarly engaged, 5 = much more engaged) | 3.60 | |

**Student feedback**. Students rated the social validity of the intervention. Figure 6 shows results of student ratings of how much they liked and found the intervention to be helpful. On average, they rated working as a team, setting goals, and earning points and stickers between 3.5 and 4, indicating that they liked those aspects of the intervention. Average ratings for answering questions and making connection were between 3.0 and 3.5, indicating that they found those aspects of the intervention to be at least somewhat helpful. There appeared to be a slight preference for answering questions after, rather than during, reading.

**Figure 6.** ***Student Ratings***

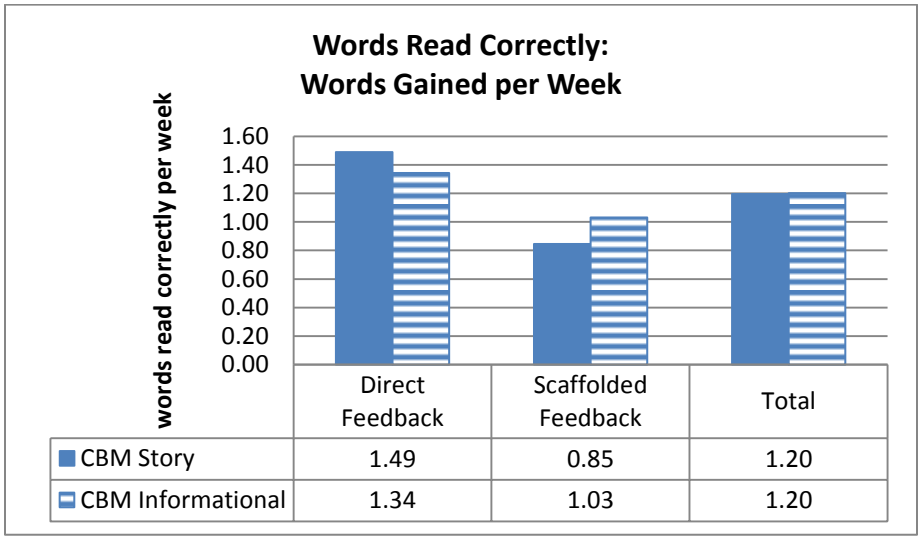



***Note.*** 1=disliked a lot/very unhelpful; 4=liked a lot/very helpful.

**Research Question 2**:
Does the effect of intervention vary based on text type (narrative stories vs. narrative informational) and feedback approach (direct vs. scaffolded)?

Currently, we are completing coding of our primary dependent variables (from the recall measures described above). Thus, results are not yet available to answer Research Question 2. Figure 7 shows the comparative gains made by students in the Direct and Scaffolded feedback conditions on both the story and informational texts. Overall, students made a gain of 1.2 words read correctly each week, with greater gains made in the Direct feedback condition than in the Scaffolded feedback condition. We are refraining from interpretation of these results until analyses are complete.

**Figure 7.** ***Words Read Correctly: Weekly Gains***




| | Direct Feedback | Scaffolded Feedback | Total |
|---|---|---|---|
| CBM Story | 1.49 | 0.85 | 1.20 |
| CBM Informational | 1.34 | 1.03 | 1.20 |

**Research Question 3:**
Do subgroups of struggling readers (paraphrasers and elaborators) respond differentially to intervention?

As indicated above, coding is still in progress. Analyses to address this question will be conducted when coding is complete.

## Round 10 (Spring Trial Run)

The purpose of the Spring Trial Run was to release responsibility to students for making connections, as well as to continue to find a way to provide students with a better understanding of the purpose of the intervention. The following variations were made to the intervention (along with rationale):

**Removing questioning prompts and sentence frames**. Question prompts and sentence frames were removed because when students are reading in other settings typically, there will not be someone to prompt them with specific questions about the text. Therefore, in the Spring Trial Run the tutor told them where to stop and make a connection but did not provide a leading question or a sentence frame. Students were taught to connect the sentence prior to the stopping point to other important sentences in the text. This required the students to consider the important components of the sentence(s) instead of the tutor providing this information for them, thus providing a release of responsibility to the students.

**Requiring connections to be stated in their own words**. In the Spring Pilot, students were encouraged but not required to provide their connections in their own words. In the Spring Pilot, frequently students were simply restating the text. It was unclear if the students were actually understanding their connections or simply restating connecting sentences that had been provided to them. Therefore, in the Spring Trial Run students were taught and required to synthesize the sentences to make an accurate connection in their own words. This further released responsibility to the students because it required students to synthesize the information to make connections in their own terms.

**A causal network and highly connected sentences visual to retell main points of the text was added**. The ultimate goal of reading is to have a coherent representation of the text. Therefore, in the Spring Trial Run, tutors further explained the purpose of the intervention with a visual representation of the causal network (Appendix Z), and used a list of the highly connected sentences in the text (Appendix ZA) to retell a coherent representation of the text. This further releases responsibility because students have a better understanding of why it is important to stop and make connections, and theoretically may be more inclined to do so while reading independently (although this was not measured).

### Participants

After the Spring Pilot, we identified two schools (one in each district) that represented demographically diverse students to participate in the Spring Trial Run. Within each school, there was one group of 3 students and one group of 4 students, for a total of 14 students who participated in the trial.

**Measure**

Once again, tutors completed daily tutoring logs (in Appendix ZB). They recorded student attendance, the total number of minutes for each session, the number of questions answers, and the number of questions requiring modeling. Then, they answered open-ended questions about the effectiveness of the session. Tutors indicated what did and did not go well in the session in terms of ease, clarity and student comprehension and provided any recommended changes.

Observations were taken of the six sessions, whole intervention, by a GRA for one of three small groups. Each session was audio recorded in order to catch how new components work in detail. The GRA took observation notes in terms of time spent, structure of the intervention, and teacher and student interaction.

**Procedures**

Tutors met with their students three times per week, for up to 30 min per session, for two weeks. In sessions 1-4, they re-read the narrative text, "A Different Aladdin." During all sessions, the stopping points to make connections were realigned to correspond to the highly connected sentences in the text. In some instances, this was the same stopping sentence but in most cases, this required the students to stop at different locations than in the original intervention.

In the first session, the students were taught to make connections without being provided a question or a sentence frame. The students were first welcomed back to the intervention and a brief explanation of the causal network for "A Different Aladdin" was provided. Students were then taught how to find only the important information in the sentence at the sentence before the stopping point and connect it to other important sentences. The students were provided a generic question of "How does this sentence connect to other important parts of the text?" Students were provided a sentence frame only when the students required additional support to answer the question. Students were encouraged to provide connections using their own words but were not required to during the first session.

The second and third session built on the first session. Students continued to make connections without a specific question, and without a sentence frame using "A Different Aladdin". However, students were also taught to examine the sentence they stopped at and the connecting sentence(s) and determine the relevant information. Students were then taught to use this relevant information to make connections using their own words.

In the fourth session, the students read, stopped at the highly connected sentences, and made connections to other important parts of the text using their own words with an unfamiliar text. The students read the entire story of "Paper Bag Princess." The students were again presented the visual of the causal network and reminded how important the highly connected sentences are to understanding a text. The rest of the session was identical to the third session but with an unfamiliar text.

The fifth and sixth sessions transitioned from making connections to using the connections to have an accurate representation of the text. In the fifth session, the students were provided modeling of how to use the highly connected sentences in "A Different Aladdin" to create an accurate, clear and full representation of the text. The students then did the same thing with "Paper Bag Princess" as a group, and in partners. Some students used a checklist of the highly connected sentences with their partner to monitor their partner to make sure he or she was stating all the important components of the text in his or her retell.

In the sixth session, the students were required to put everything together using two new texts. The students read following the normal procedure without question or sentence prompts.

The students stopped and made connections at the highly connected sentences in their own words and discussed why each connection was important to the understanding of the overall story. Then they used the list of highly connected sentences provided to them to retell the important parts of the text while adding relevant details. Some students again used a checklist to monitor their partners retell and provide feedback. Finally, the students reviewed everything they had learned.

Tutors brought completed logs to weekly meetings to facilitate discussion about how the intervention was going. Each tutor described how the week's sessions had gone, including how well they thought students responded to each session, how well they thought individual students were responding to the intervention, and their overall impressions of the Spring Trial Run. The PI and Project Coordinators took extensive notes during these discussions.

**Results**

General feedback included the following. This feedback, along with more extensive notes in tutoring logs and meeting minutes, will be used to inform further revisions to the intervention.

**Removing question prompts and sentence frames**. Overall, the students did not seem to struggle with this. The students, once taught, understood how to connect sentence to sentence without a question or a sentence frame.

**Requiring connections to be stated in their own words**. Students initially struggled with being able to do this independently but by the fourth session students seemed to be able to state connections in their own words. However, there were a few students who still struggled with putting connections into their own words throughout the Spring Trial Run.

**A causal network and highly connected sentences visual to retell main points of the text was added**. Some students found the causal network confusing, while others found it helpful. However, most students understood its purpose. Students seemed to understand the highly connected sentences visual and were able to use it to retell the text by the end of the six sessions.

## C. Opportunities for Training and Professional Development

Opportunities for training/professional development (PD) to GRAs/URAs and teachers/specialists are describe in Table 17. PD opportunities include Knowledge Building activities, in which we provided opportunities for research staff and teachers to learn about important aspects of the grant, Technical Training, in which we provided research staff with training to administer tests and intervention (much of which involves relevant knowledge and skills that are generalizable to their educational careers), and Mentoring (in which project faculty work with individual graduate students to develop their own research questions and analyses using project data).

**Table 17.** ***Professional Development Opportunities***

| Type of PD | Content (Provided By) | Provided To |
|---|---|---|
| Knowledge Building | Overview/rationale for grant, theoretical and empirical underpinnings (McMaster, Janda, Pinto) | Research Team (in Lab Meetings), Teachers (in Focus Groups) |
| | Research design (McMaster) | Research Team (in Lab Meetings) |
| Technical Training | Human subjects and responsible conduct of research training (University of MN) | All research staff |
| | Selecting texts and developing questions (McMaster) | GRAs |
| | Administering and scoring CBM maze and oral reading (Pinto) | GRAs and URAs |
| | Administering think-aloud (Janda & Pinto) | GRAs and URAs |
| | Coding think-aloud (Janda) | GRAs and URAs |
| | Administering intervention (McMaster, Pinto, Janda, Bethel, Oberg) | GRAs |
| | Administering and coding recall task (McMaster, Janda, Pinto, van den Broek, & Espin) | GRAs |
| Mentoring | Participating in applied research (included writing a reflection on the experience) | URAs |
| | Writing research reports | GRAs |
| | Developing independent research questions and data analysis (McMaster, van den Broek, Espin) | GRAs |

## D. Results Disseminated to Communities of Interest

Results of the project have been shared to the following communities of interest (this list is cumulative for Years 1 & 2):

(1) University of Minnesota College of Education and Human Development: Poster Presented at annual CEHD Research Day (March 20, 2012)
(2) University of Minnesota Special Education Program: Brown Bag Colloquium for Special Education faculty and students (April 4th, 2012)
(3) Participating districts: Year 1 Annual Report.
(4) Participating teachers: Making the Right Connections Teacher Share (small group meetings with teacher participants throughout the month of May, Years 1 & 2, following completion of the Spring Pilot)
(5) National special education community: McMaster, K. L., Espin, C., van den Broek, P., Janda, B., Pinto, V., & Jung, P. G. (2013, April). Making the right connections: Improving the comprehension of struggling readers. Presented at the 2013 Annual Convention of the Council for Exceptional Children.
(6) National educational research community: McMaster, K. L., Pinto, V., van den Broek, P., Espin, C., & Janda, B. (2013, February). Making the right connections: Improving the comprehension of struggling readers. Presented at the 2013 Pacific Coast Research Conference, San Diego, CA.

We also plan to share an abbreviated version of the Year 2 report with our two participating districts, and will offer to meet with interested school personnel to present our Year 2 findings. Additional conference proposals for posters and presentations are also underway.

## E. Accomplishing Goals in the Next Reporting Period

In the next reporting period, we will complete Year 2 analysis, finalize revisions to the current version of the intervention, and prepare presentations and papers for dissemination (on schedule with original plan). In addition, we will address specific Year 2 goals, with no major changes to the goals outlined in the performance agreement.

## II. PRODUCTS

### A. Publications and Presentations

**Presentations (cumulative for Years 1 & 2)**

McMaster, K., Pinto, V., Janda, B., van den Broek, P., & Espin, C. (March, 2012). Making the right connections to improve the comprehension of struggling readers. Poster presented at the Annual University of Minnesota CEHD Research Day, Minneapolis, MN.

McMaster, K., Janda, B., & Pinto, V. (April, 2012). Under the big top: Balancing theory, evidence, and practice to develop a reading intervention. Presented to the monthly University of Minnesota Special Education Program Brown Bag Colloquium, Minneapolis, MN.

McMaster, K. L., Pinto, V., van den Broek, P., Espin, C., & Janda, B. (2013, February). Making the right connections: Improving the comprehension of struggling readers. Presented at the 2013 Pacific Coast Research Conference, San Diego, CA.

McMaster, K. L., Espin, C., van den Broek, P., Janda, B., Pinto, V., & Jung, P. G. (2013, April). Making the right connections: Improving the comprehension of struggling readers. Presented at the 2013 Annual Convention of the Council for Exceptional Children.

**Publications (submitted and in progress)**

McMaster, K. L., van den Broek, P., Espin, C. A., Pinto, V., Janda, B., Lam, E. A., Hsu, H., Jung, P., van Boekel, M., & Bethel, A. (submitted). Developing a reading comprehension intervention: Translating cognitive theory to educational practice. Contemporary Educational Psychology.

McMaster, K. L., Espin, C. A., & van den Broek, P. (in progress). Making connections: Linking cognitive theory and educational intervention to improve reading comprehension. Invited article for Learning Disabilities Research and Practice.

### B. Website

http://www.cehd.umn.edu/EdPsych/readinglab/

### C. Other Products

Measures of fit, feasibility, and promise of the intervention (referenced as Appendices A-Z in the report) have been developed and are attached.

## III. PARTICIPANTS AND OTHER COLLABORATING ORGANIZATIONS

### A. What individuals have worked on the project?

| Name | Project Role | Person mo. | Contribution | Funding Support | Foreign Country |
|---|---|---|---|---|---|
| Kristen McMaster | Principal Investigator | 3 | Oversees all aspects of project | IES | NA |
| Paul van den Broek | Co-PI | 2.4 | Provides leadership in Discourse aspects of project | IES | Leiden University, The Netherlands |
| Christine Espin | Co-PI | 1.2 | Provides leadership in Intervention aspects of project | IES | Leiden University, The Netherlands |
| Viveca Pinto | Project Coordinator | 6 | Coordinates intervention activities | IES | NA |
| Bonita Janda | Project Coordinator | 6 | Coordinates discourse activities | IES | NA |
| David Rapp | Discourse Expert | (8 hours) | Consulted on texts and question development | IES | NA |
| Panayiota Kendeou | Discourse Expert | (8 hours) | Consulted on Think-Aloud analyses | IES | Neapolis University, Cyprus |
| Joseph Jenkins | Intervention Expert | (8 hours) | Consulted on intervention development | IES | NA |
| Stanley Deno | Intervention Expert | (8 hours) | Consulted on intervention development | IES | NA |
| Amy Bethel | Graduate Research Assistant | 3 | Data collection & analysis, intervention development & implementation, report writing | IES | NA |
| Pyung-Gang Jung | Graduate Research Assistant | 6 | Data collection & analysis, intervention development & implementation, report writing | IES | NA |
| Hui-Chen Hsu | Graduate Research Assistant | 6 | Data collection & analysis, intervention development & implementation, report writing | IES | NA |
| Elizabeth Oberg | Graduate Research Assistant | 2.25 | Data collection & analysis, intervention development & implementation, report writing | IES | NA |
| Jaehyun Shin | Graduate Research Assistant | 3 | Data collection & analysis, intervention development & implementation | Dept. funds, IES | NA |
| Michael Herriges | Graduate Research Assistant | 1.125 | Data collection & analysis, intervention development & implementation | IES | NA |
| Katherine Brodeur | Graduate Research Assistant | 1.125 | Data collection & analysis, intervention development & implementation | IES | NA |
| Martin van Boekel | Graduate Research Assistant | 3 | Data collection & analysis, intervention development & implementation, report writing | IES | NA |

In addition, Undergraduate Research Assistants (URAs; junior or senior Psychology majors) worked on the project as volunteers or for course credit for varying numbers of hours.

- Aaron Hauger (spring only)
- Neil Holmberg (spring only)
- Leonardo Lee (fall only)
- Katlin Malm (fall and spring)
- Katherine Melton (spring only)
- Allison Morrow (fall and spring)
- Annie Moynihan (spring only)
- Nasima Omar (fall only)
- Ally Wilcox (fall only)
- Nicole Woodall (fall and spring)

**What other organizations have been involved as partners?**
Bloomington Public Schools and Minneapolis Public Schools served as the district sites for all school-based activities described in this report.

**Have other collaborators or contacts been involved?** No.

## IV. IMPACT

### A. Impact on the development of the principal disciplines of the project

The current project has the potential to impact the principal disciplines (cognitive science, reading research) in the following ways:

1. We are shedding light on how cognitive models of reading can be translated into reading intervention for struggling readers. Particularly, we are (a) examining whether cognitive theories about the importance of causal connections can be translated into a questioning intervention, and (b) determining whether subgroups of struggling readers (identified using cognitive processing methodologies) are differentially responsive to different approaches to reading intervention. In turn, results may help refine cognitive models of reading. For example, our findings suggest that students who struggle with reading *are able* to make text-based, causal connections; however, they do not necessarily do so unless they are provided with specific questions and prompts.

2. We are contrasting two approaches to providing feedback: a direct approach versus a scaffolded approach. Findings may inform the field about the most efficient and promising way to individualize intervention in a small-group context.

3. We are determining critical features that increase the feasibility of implementing reading comprehension intervention in school settings, including group size (no fewer than 3-4 students), frequency and duration (3 times per week for 20-30 min), interventionist (tutor experienced with children who has substantial training and support, but does not necessarily need to be a licensed teacher), highly scripted and well organized materials packaged for immediate use, development of texts and questions that fit both the intervention and schools' instructional programming.

### B. Impact on other disciplines

None at this time.

### C. Impact on the development of human resources

This project has provided opportunities for graduate and undergraduate research assistance and teachers to learn about cognitive models of reading, intervention design, research design, and to gain experience implementing a variety of assessments and intervention protocols.

### D. Impact on resources that form infrastructure

None at this time.

### E. Impact on technology transfer

None at this time.

### F. Impact on society

None at this time.

### G. Dollar amount spent in foreign countries

$94,215.35 in the current reporting period (subcontract with Leiden University, in The Netherlands).

## V. CHANGES/PROBLEMS

There have been no changes to the activities outlined in the Performance Agreement (as revised on 8/17/12), and no problems in the current reporting period. No changes or problems are anticipated for the upcoming reporting period.

## VI. BUDGETARY INFORMATION

**Budget Justification**

**A. Key Personnel**

***Kristen McMaster, Principal Investigator****,* (2.25 Cal. Months AY/.75 summer)
Dr. McMaster has devoted 25% FTE (2.25 cal months AY and .75 cal months summer) to this project. She oversees all major grant activities, including management of the budget; supervision of Site and Intervention coordinators, GRAs, and data collectors; development and revisions of intervention materials and procedures; conducting training of interventionists and teachers; data collection from various sources (experts, teachers, pilot studies); designing pilot studies; data analysis and interpretation. She also prepares reports required by the agency.

**B. Other Personnel**

***Bonita Janda, Discourse Coordinator*** (6 cal months over 12 months)
Ms. Janda devotes 50% FTE (6 cal months over 12 months) to this project. She provides technical assistance for this project under the direct supervision of the PI. Specifically, she supervises GRAs and URAs; participates in development and revisions of intervention materials and procedures; coordinates data collection efforts with experts; coordinates scheduling for screening and other assessments; coordinates scheduling for interventions conducted during trial runs and pilot studies; oversees data transcription, coding, and entry; assists with data analysis and interpretation; and assists with writing reports.

***Viveca Pinto, Intervention Coordinator*** (6 cal months over 12 months)
Dr. Pinto devotes 50% FTE (6 cal months over 12 months) to this project. She began the project on a graduate research assistant salary base, and switched to a research fellow salary base upon completing her doctorate in February, 2012. Dr. Pinto coordinates development and revisions of intervention materials and procedures; coordinates data collection efforts with teachers, including facilitating focus groups; coordinates and participates in training of interventionists and teachers; conducts interventions and observations during intervention trial runs and pilot studies; assists with data analysis and interpretation; and assists with writing reports.

***Graduate Research Assistants*** assist with development and revisions of intervention materials and procedures; administer and score screening measures and other assessments; serve as interventionists; code Think-Aloud protocols and data obtained from questionnaires, surveys, and observations; attend meetings as needed; and assist with data entry and analysis.

***Hourly Data Collectors*** (originally budgeted for 3.72 cal months over 12 months)
We moved the hourly data collector budget line into the Graduate Research Assistant line. Having GRAs dedicated to the project better suited project needs than would hourly data collectors, given the amount of training and time needed to carry out project data collection activities.

**D. Travel**

***Conference Travel and Mileage*** (budgeted = $5,000 per year, Year 2 expense = $5090.99)
Included travel to two conferences to present Year 1 findings, and mileage for project staff to travel to schools.

**E. Other Direct Expenses**

***Materials and Supplies*** (budgeted=$4,300 Yr. 2, actual=$1722.04.)
These funds were used for supplies necessary for project implementation (e.g., intervention binders and folders). An iPad was purchased for onsite data collection and note taking. Supply expenses have not been as high as anticipated, as it was not necessary to purchase assessments and software that were available through other resources in the department.

***Transcription Services*** (budgeted=$2000 Yr. 2, actual=$546.00)
We were able to complete some of the transcription internally, so expenses were not as high as anticipated.

***Consulting*** (budgeted = $6,000/year, actual Yr. 2 = $4,000)
Funds were requested to pay each of the following consultants $500/day X up to 2 days each per project period=$1,000.

Dr. Stan Deno, University of Minnesota (emeritus)
Dr. Joe Jenkins, University of Washington
Dr. Panayiota Kendeou, McGill University
Dr. David Rapp, Northwestern University

We have worked with all consultants except for Drs. Graesser & Fuchs, who fulfilled their requested role during Year 1 of the project. Our plan is to solicit additional feedback from them at the end of Year 2 when final revisions are made based on Year 2 results.

***Subaward*** (budgeted=$103,418 Yr. 2; actual = $94,215.35)
Drs. Paul van de Broek and Christine Espin of Leiden University serve as co-investigators on the project. We are awaiting an additional Year 2 invoice for the remaining costs.

***Teacher Stipends and Substitute Costs*** (budgeted = $24,320 Yr. 2, actual = $14,865.45)
Funds were used to pay teacher stipends and reimburse substitutes needed for focus groups. Substitute costs were not as high as anticipated because 4 teachers did not attend focus groups during school hours (other arrangements were made to ensure they had input), and several of the specialists did not need substitute coverage while they were at the focus group sessions.

***Copying, Background Checks, Student Incentives*** (budget = $5,325 Yr. 2, actual = $1,330.11)
Funds were used to purchase gel pens and stickers to be used as incentives for intervention. Costs were not as high as expected as we were able to find discounted items for student incentives.

## F. Indirect Costs

The University of Minnesota applies a 34% Indirect Cost rate to all direct costs minus graduate research assistant fringe and the amount of the subcontract over $25,000. The on-campus indirect cost rate is being applied given that more than 50% of personnel on this project are housed on-campus.

**Changes to the Budget**
There were no changes to the budget that resulted from modifications to project activities.
There were no changes to the budget that affected our ability to accomplish project activities as described in the performance agreement

We do not anticipate that funds will be carried over to the next project period. The current balance of $82,049.14 should be spent down by August 31, 2013, because July-August costs are not reflected in this report.

There are no anticipated changes in the budget for the next project period.